

TOSHIBA

TOSHIBA Portable Printer

B-FP Series

ESC/POS Specification

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TOSHIBA TEC CORPORATION

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1. SCOPE

This specification applies to the ESC/POS for use with the B-FP series.

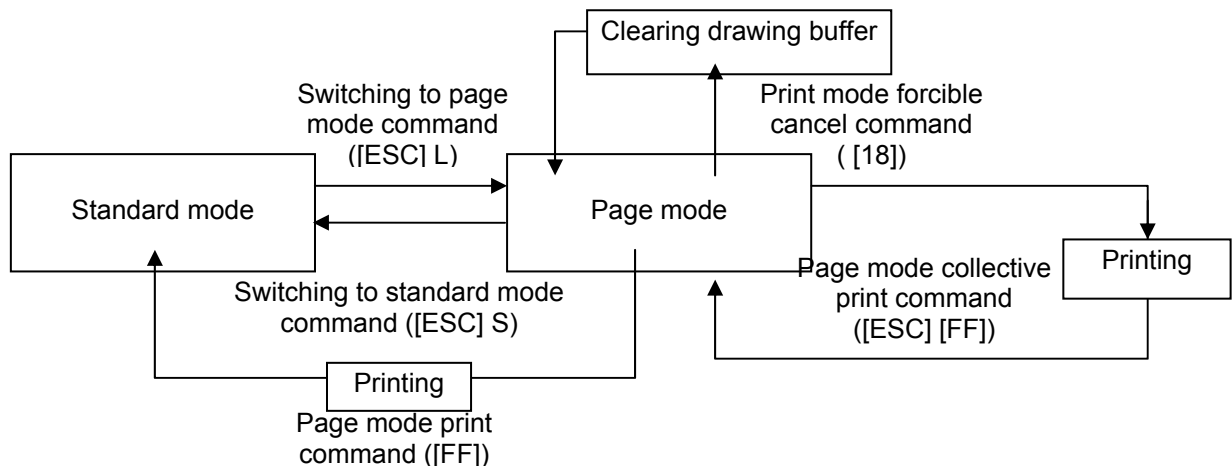
2. GENERAL DESCRIPTION

This printer has 2 modes: standard mode and page mode.

In standard mode, the printer performs a print job or feeds paper every time a print or feed command is received. In page mode, however, all received print or feed commands are just expanded in the print area on the memory and the printer does not operate. Once the Page Mode Collective Print Command ([ESC] [FF]) or Page Mode Print Command ([FF]) is executed, the printer prints all data expanded in the print area in a collective manner.

For instance, if printing and feeding of data "ABCDEFGH" [LF] is performed, "ABCDEFGH" will be printed, and a new line will be started in standard mode. In page mode, "ABCDEFGH" will be written into the specified print area on the memory, and the memory position to write subsequent print data will be shifted 1 line.

The printer is placed in page mode upon the receipt of the Switching to Page Mode Command ([ESC] L). Any commands received subsequently are processed for page mode. Executing the Page Mode Collective Print Command ([ESC] [FF]) allows all received data to be printed in a collective manner. In addition, executing the Page Mode Print Command ([FF]) allows all received data to be printed in a collective manner, and returns the printer to standard mode. Executing the Switching to Standard Mode Command ([ESC] S) returns the printer to standard mode without printing the print data for page mode. Note that the print data is cleared.



Either of the language types, Kanji, Chinese and Korean, can be implemented. Any font other than the on-board fonts is selectable.

3. CONSTRUCTION

PRINTER Firmware

Compatible printer firmware is required. Update the firmware as necessary.

For how to update it, refer to the "Program Download Operation Specification."

4. START OF ESC/POS

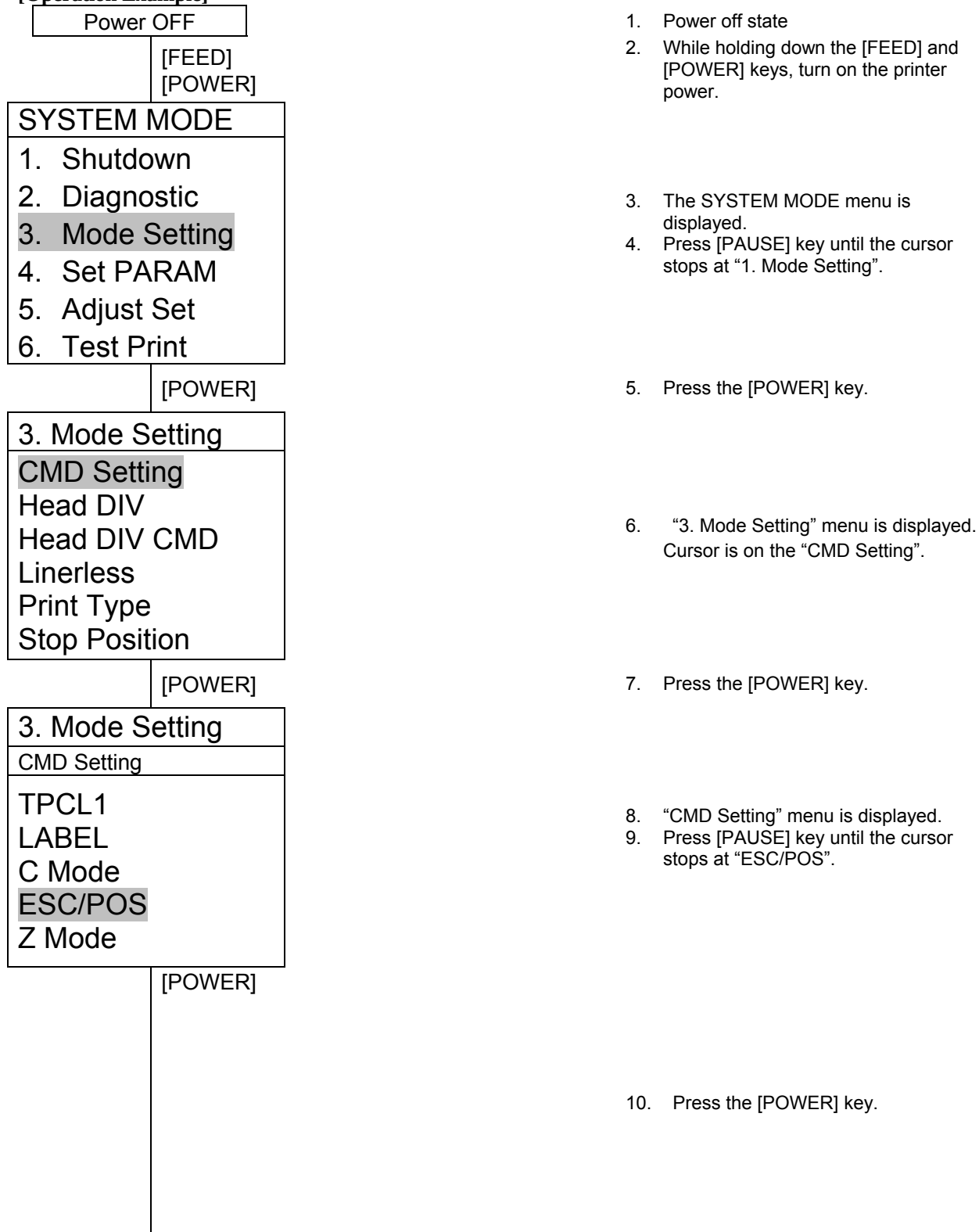
Place the printer in ESC/POS Mode to use the ESC/POS Mode.

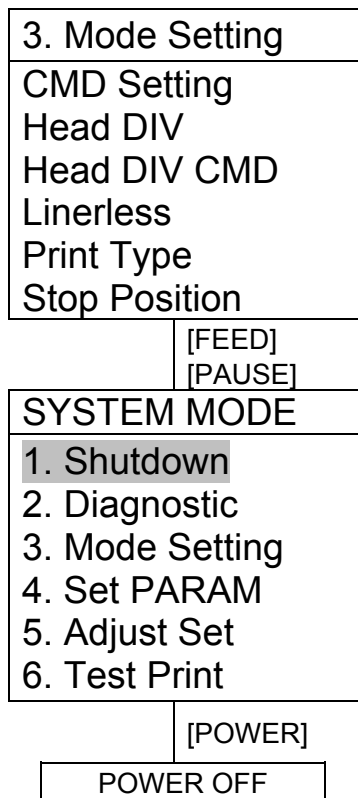
Refer to the subsequent pages for the operation example on the B-FP. (Also refer to the “Key Operation Specification”.)

NOTES

1. Enabling or disabling the ESC/POS Mode is also possible by using the BCP setting tool.

[Operation Example]





11. Menu is returned to “3. Mode Setting” menu.
12. Press the [FEED] and [PAUSE] key.
13. Menu is returned to “SYSTEM MODE” menu.
14. Press [FEED] key until the cursor stops at “1. Shutdown”.
15. Press the [POWER] key.
16. Power off state

5. ESC/POS COMMANDS

5.1 Outline of command

5.1.1 Format of Interface command

ESC	Command
GS	Command
FS	Command
US	Command
HT	
LF	
FF	
CAN	

Control codes are as listed below:

- ESC (1BH)
- GS (1DH)
- FS (1CH)
- US (1FH)
- HT (09H)
- LF (0AH)
- FF (0CH)
- CAN (18H)

5.1.2 How to use reference

Function	Describes the outline of the function of the command.
----------	---

Format	Shows the format of the command.
--------	----------------------------------

The format designation method should conform to the following rules:

- n refers to a parameter item.
- Brackets and parentheses (e.g. [] or < >) are used only for description, and must not be transmitted in practice.
- Other symbols must always be inserted at designated positions before being transmitted.

Term	Explains the term(s) used in the format.
------	--

Default value	Default value of a parameter (at power-on or switching to ESC/POS mode)
---------------	---

Explanation	Explains the command in detail.
-------------	---------------------------------

Refer to	Related commands
----------	------------------

5.1.3 precautions

Print density fine adjustment cannot be selected in the ESC/POS mode. The mode must be switched to the TPCL, LABEL or RECEIPT1 mode once, to select print density fine adjustment.

5.2 Font specifications

Font type:

Alphanumeric characters: 234

Extended graphics: 128
characters × 4

Chinese characters: JIS 1 level 2

		ANK	Chinese character, Chinese, or Korean
FONT SIZE 1	Font A	12×24	24×24
	Font B	9×24	
FONT SIZE 2	Font A	13×28	
	Font B	10×28	

5.3 External (custom) characters specifications

- ANK

Font A 234 characters (20h to FEh)

Font B 234 characters (20h to FEh)

- Chinese characters

* 7721h to 777Eh for JIS, EC40h to EC9Eh (except for EC7Fh) for shift JIS, and F7A1h to F7Feh for the EUC code

- Chinese

FF40h to FF7Eh, FF80h to FFFCh

- Korean

FF40h to FF7Eh, FF80h to FFFCh

* The specified writable characters are not backed up. (They apply until the printer is turned off.)

5.4 Input data buffer

Input data buffer (receive buffer) size: 512 Kbytes

5.5 Value specified by each command

The values specified by each command are commonly applied in standard and page modes. However, as for the following commands, values must be independently specified and maintained in standard and page modes.

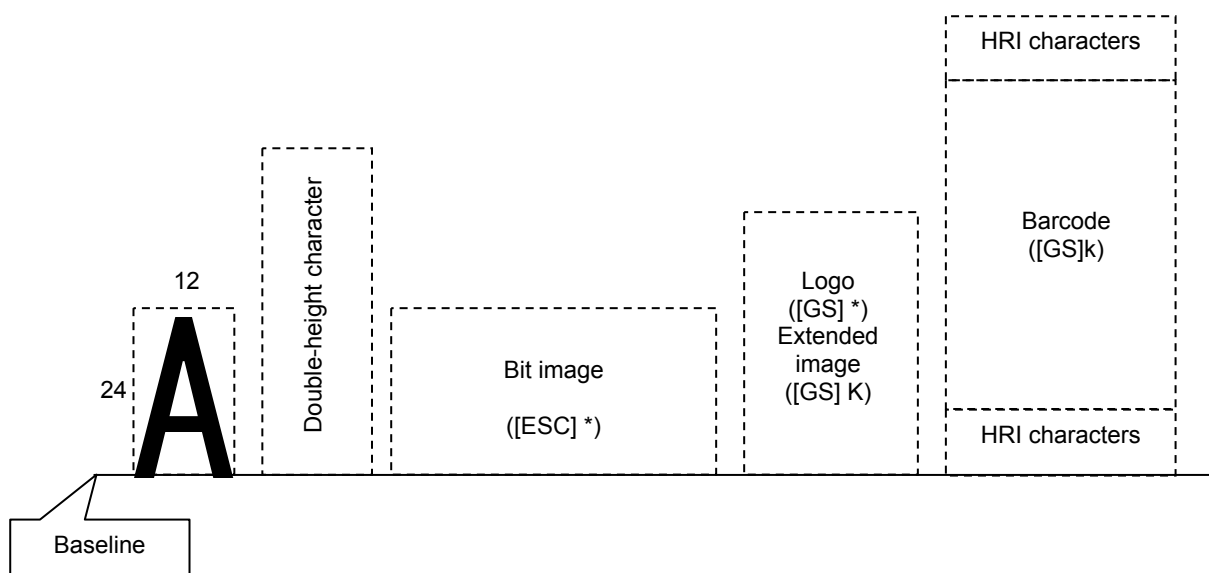
- [ESC] [SP] : Setting of spacing on the right of a character
- [FS] S : Setting of spacing on both sides of a chinese character

5.6 Expansion of print data in print area

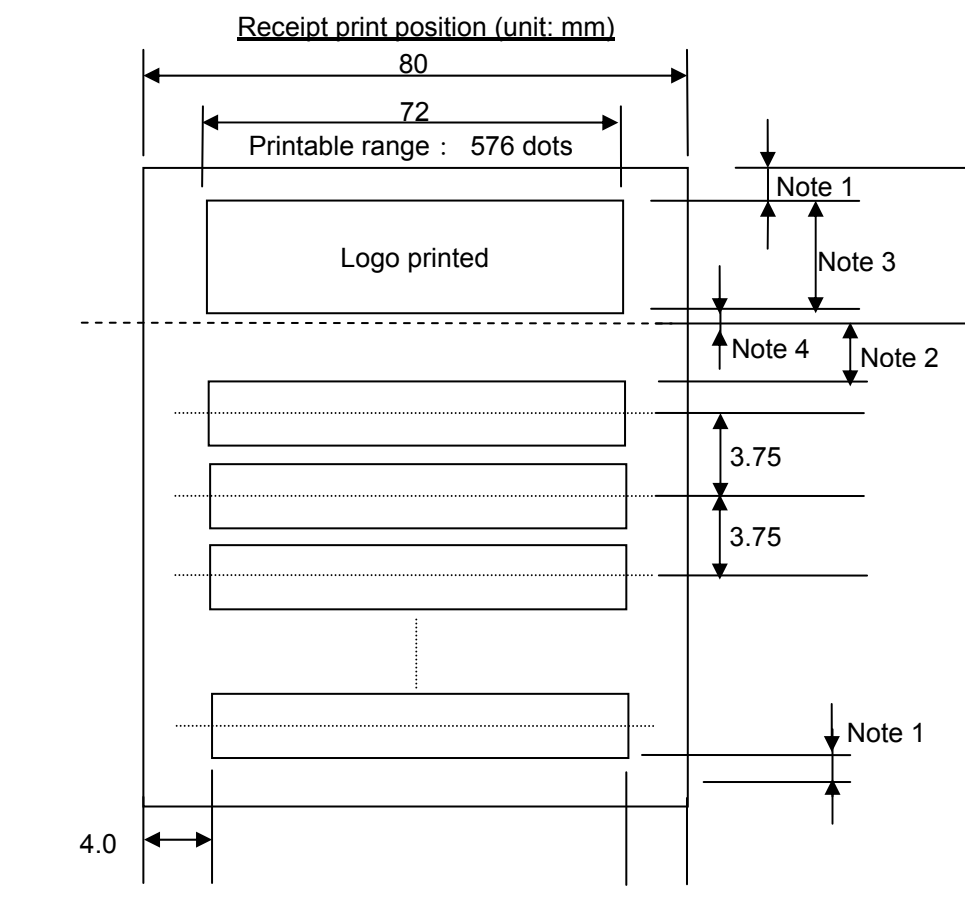
The print data is expanded in the print area as follows.

- (1) The print area is defined by the Page Mode Print Area Set command ([ESC] W). The left end of the printer as one face must be considered as the starting point of the print area (x0,y0) at the moment when printing or feeding before the printer receives [ESC] W is completed. The print area is formed by a rectangle that is dx pitches wide (from the starting point in the x or horizontal direction) and dy pitches high (from the starting point in the y or vertical direction). (If the print area is not defined by [ESC] W, a default value will be applied.)
- (2) If the printer receives print data after the print area is defined by the Page Mode Print Area Set command ([ESC] W) and the print direction is defined by the Page Mode Print Direction Set Command ([ESC] T), characters and downloaded bit images are expanded aligning their bottom left point to the baseline.
- (3) If the print data (including space on the right of a character) deviates from the print area before a command that includes line feed (Line Feed Command ([LF]) or Feed Length Set Command ([ESC] J) is received, line feed will be performed automatically within the print area. The position where print data is expanded is shifted 1 line. Also, the next expansion position is the beginning of the line. Line feed must be performed in accordance with the line feed length specified by the 34-dot Line Feed Length Set Command ([ESC] 2) or Minimum Dots for Line Feed Length Set Command ([ESC] 3).
- (4) The default line feed length is 27 dots. Therefore, if the print data for the next line includes double-height characters or downloaded bit image, the default line feed length may be insufficient. As a result, the higher order dots may be superimposed on the previous print data. Therefore, the line feed length must be increased.
- (5) Printing of a barcode in horizontal orientation is not allowed due to a large reading error.

[Position where print data is expanded]



5.7 example of receipt printer print layout (B-FP3D-GH/GS)



- Note 1) A minimum margin of 2 mm must be reserved at the top and bottom of a receipt.
- Note 2) If printing starts from a stand-by position (or motor stop position) after a logo is printed, a blank must be inserted for a minimum of 24 pulses (3.0 mm) (or 48 pulses (6.0 mm) or over recommended). (This is because of reserving a slow-up area. The number of slow-up pulses varies depending on the print speed.)
- Note 3) A logo must be printed within the range of 21 mm at maximum.
- Note 4) After a logo is printed, a blank must be inserted for a minimum of 24 pulses (3.0 mm) (or 48 pulses (6.0 mm) recommended).

5.8 Exception handling

5.8.1 Undefined code

This applies to the codes that fall in the range between 00h and 1Fh in the character code table. If a code that is not defined in the range is inputted as a command, 1 byte must be discarded, and the subsequent data must be processed as normal data.

<Ex.> Data string [30H][31H][03H][32H][0AH][33H] is inputted.

[30H] [31H] [03H] [32H] [0AH] [33H]

- This is defined as line feed command [LF]

This is undefined code, so ignored.

Data string actually processed



[30H] [31H] [32H] [0AH] [33H]

5.8.2 Undefined command

If a code following the [ESC] (1Bh), [FS] (1Ch), or [GS] (1Dh) is not defined as a command, [ESC], [FS], or [GS] and the following code (2 bytes in total) must be discarded.

<Ex.> Data string [30H][1BH][22H][31H][32H] is inputted.

[30H] [1BH] [22H] [31H] [32H]

- [22H] is undefined code, so ignored.

Data string actually processed



[30H] [31H] [32H]

5.8.3 Setting out of defined range

If a numeric value outside the defined range is inputted, the command must be ignored, and an already set value must not be changed. For commands that have multiple arguments, once a numeric value out of the defined range is inputted, command processing is aborted, and the subsequent data is processed as normal data.

<Ex.> Data string [1BH][2DH][05H] is inputted.

[1BH] [2DH] [05H]

- This is undefined code in Underlining Set command [ESC]-.

- This is defined as Underlining Set command [ESC]-.

Data string actually processed



[05H] is out of the defined range. Therefore, the data string must be discarded. The already set underline remains unchanged.

6. SUPPORTED COMMANDS

6.1 Horizontal Tab command [HT]

Function	Moves a print position to the next horizontal tab position.
Format	ASCII: HT Hex: 09h
Default	Printer initialization sets 32 tabs at 9, 17, 25... (Every 8 digits)
Explanation	(1) This command will be ignored if the next horizontal tab position is not specified. (2) The Horizontal Tab Positions Set Command ([ESD] D) must be used to specify a position of a horizontal tab. [Program example] <pre>fprintf(stdprn, "01234567890123456789012345678901234\n"); for(i=0; i<4; i++) { fputc('\t', stdprn); /* Horizontal tab */ fputc('H', stdprn); } fputc('\n', stdprn);</pre> [Print example]: Default horizontal tab position 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 H H H H
Reference	Horizontal Tab Positions Set Command ([ESD]D)

6.2 Print and carriage return [CR]

Function	Prints one line from the buffer and feed paper one line.
Format	[CR]
Term	ASCII: [CR] Hex: 0Dh
Explanation	(1) The printer can be set through the printer setting command to ignore or use this command.

6.3 line feed command [LF]

Function	Prints data and inserts a line feed.
----------	--------------------------------------

Format	ASCII: LF Hex: 0Ah
--------	-----------------------

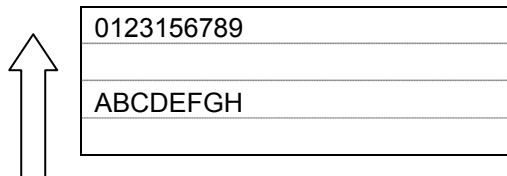
- | | |
|-------------|--|
| Explanation | <ul style="list-style-type: none">(1) If print data is received with the specified number of digits per line exceeded and no line feed ([LF]), it will be printed with a line feed inserted automatically.(2) If this command is received without print data, a line feed of a specified length will be just inserted.(3) Data that contains 4,096 bytes in a single line (up to [LF]) must not be transmitted. If the number of bytes exceeds 4,096, the correct functioning of the printer will not be guaranteed. |
|-------------|--|

[Program example]

```
fprintf( stdprn, "0123456789\n");  
fputc( '\n', stdprn );  
    feed */  
fprintf( stdprn, "ABCDEFGH\n");
```

/* Line

[Print example]



Reference	34-dot Line Feed Length Set Command ([ESC] 2) Minimum Dots for Line Feed Length Set Command ([ESC] 3)
-----------	--

6.4 Page Mode Print command [FF]

Function	Prints the data expanded in the print area in page mode, and returns the printer to standard mode.
Format	ASCII: FF Hex: 0Ch
Explanation	(1) This command is ignored except when page mode is selected. (2) The data that has been drawn is all erased after printing is completed. (3) The print area specified by the Page Mode Print Area Set Command ([ESC] W) is initialized. (4) After this command is executed, the next print starting position is the beginning of a line.
Reference	Page Mode Collective Print Command ([ESC] [FF]) Switching to Page Mode Command ([ESC] L) Switching to Standard Mode Command ([ESC] S)

6.5 Page Mode Cancel command [CAN]

Function	Clears all data in the currently specified print area for page mode.
Format	ASCII: CAN Hex: 18h
Explanation	(1) This command is ignored except when page mode is selected. (2) The data contained in the currently specified print area will be cleared even if it is the data contained in the previously specified print area. The print starting position in the currently specified print area is returned to the starting position defined by the Page Mode Print Area Set Command ([ESC] W).
Reference	Switching to Page Mode Command ([ESC] L) Page Mode Print Area Set Command ([ESC] W)

6.6 Page Mode Collective Print command [ESC] [FF]

Function	Prints the data expanded in the print area in page mode altogether.
Format	ASCII: ESC FF Hex: 1Bh 0Ch
Explanation	(1) This command is ignored except when page mode is selected. (2) The values specified by the Page Mode Print Direction Set Command ([ESC] T) or the Page Mode Print Area Set Command ([ESC] W) and the position where the print data is expanded are maintained after printing is completed.
Reference	Page Mode Print Command ([FF]) Switching to Page Mode Command ([ESC] L) Switching to Standard Mode Command ([ESC] S)

6.7 Spacing Set command [ESC] [SP]

Function	Specifies spacing (or the number of dots) on the right of an ANK character.
Format	[ESC] SP n
Term	ASCII: ESC SP Hex: 1Bh 20h
[Parameter]	
n: Spacing (or the number of dots) on the right $0 \leq n \leq 32$ (00h $\leq$ n $\leq$ 20h)	
Default value	n : 1
Explanation	(1) For a double-width character, spacing will also be doubled. (2) The maximum permissible spacing on the right of a character is 32 dots. If a specified value exceeds 32, the command is ignored. (3) Spacing can be independently specified in standard and page modes. (4) The value specified by this command is not applicable to Chinese characters. Instead, the Chinese Character Spacing Set Command ([FS] S) must be applied.

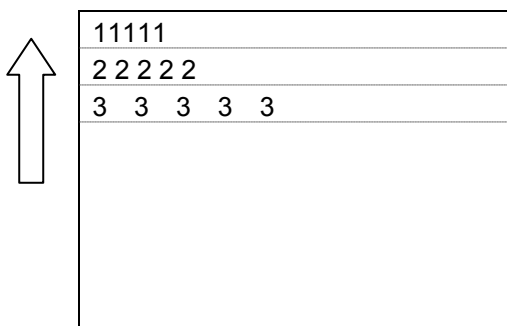
[Program example]

```

fprintf( stdprn, "\033 ");           /* 1 dot on the right */
fputc( 1, stdprn );
fprintf( stdprn, "11111\n");
fprintf( stdprn, "\033 %c", 13 );    /* 13 dots on the right */
fprintf( stdprn, "22222\n");
fprintf( stdprn, "\033 %c", 26 );    /* 26 dots on the right */
fprintf( stdprn, "33333\n");

```

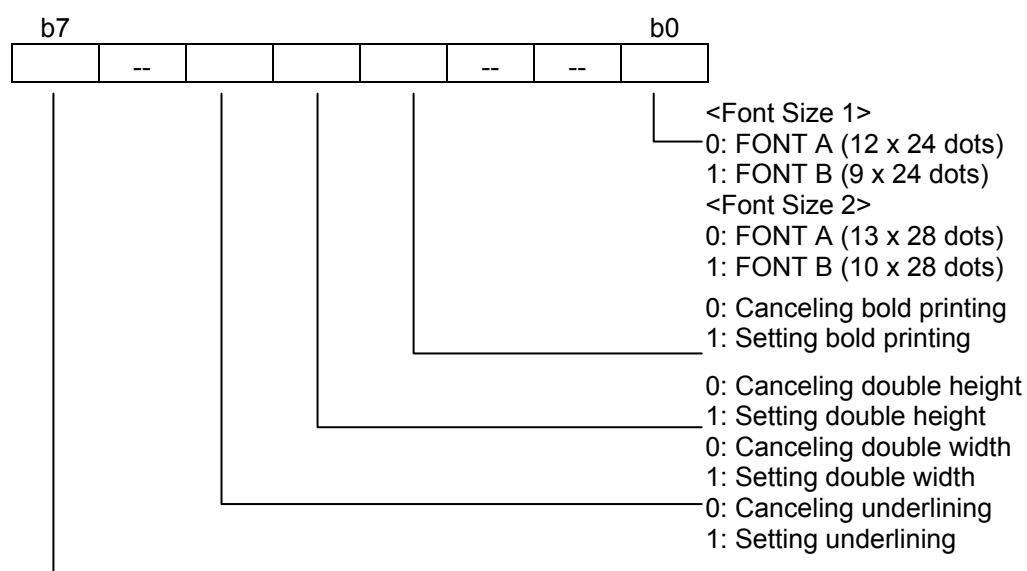
[Print example]



11111
2 2 2 2 2
3 3 3 3 3

6.8 Print Mode All Set command [ESC] !

Function	Performs character control (specifies print modes altogether).
Format	[ESC] ! n
Term	ASCII: ESC ! n Hex: 1Bh 21h n
[Parameter]	
n: Specifying print mode	$0 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh)



Default value n : 0

- Explanation
- (1) With underlining set, a portion skipped by HT or a character rotated 90 degrees clockwise will not be underlined.
 - (2) An underline is 1 dot thick. However, the thickness of an underline is determined in accordance with the thickness specified by the Underlining Set Command ([ESC]-).
 - (3) Bold printing can also be set or canceled by the Bold Printing Set Command ([ESC]E), but the most recently processed command is applied.
 - (4) An ANK character size can also be specified by the Character Size Set Command ([GS] !), but the most recently processed command is applied.
 - (5) Underlining can also be set or canceled by the Underlining Set Command ([ESC]-), but the most recently processed command is applied.
 - (6) If both double height and double width are set at the same time, a character size will be quadrupled (double height $\times$ double width).

[Program example]

```
fprintf( stdprn, "\033! ");  
width */
```

/* Normal

```
fputc( 0, stdprn );
```

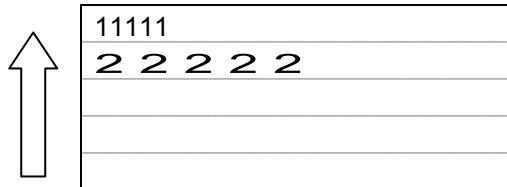
```
fprintf( stdprn, "11111\n");
```

```
fprintf( stdprn, "\033 %c", 0 × 20 );
```

/* Double width */

```
fprintf( stdprn, "22222\n");
```

[Print example]



11111
2 2 2 2 2

Reference

Underlining Set Command ([ESC] -)

Bold Printing Set Command ([ESC] E)

Character Size Set Command ([GS] !)

6.9 Downloaded Character (Writable Character) Set command [ESC] %

Function	Sets or cancels downloaded characters (writable characters).
Format	[ESC] %n
Term	ASCII: ESC % n Hex: 1Bh 25h n [Parameter] n: Code Page and Character set Accepted Value: 00 to FF 00 : Code Page 437 and internal ANK characters set 02 : Code Page 850 and internal ANK characters set Other value: Only Character table can be set. Code page is set previous value. only LSB is judged. In case LSB is 0, Code page is previous value. LSB : 0 internal ANK Character LBS : 1 User-defined character [for example] (1) In case of FF 11111111 User-defined character (2) In case of 3C 00111100 internal ANK character
Default value	n: 00
Explanation	(1) If downloaded characters (writable characters) are not registered with a specified code, the internal ANK characters will be printed. Also, when the downloaded characters (writable characters) are canceled, the internal ANK characters will be selected automatically for printing.
Reference	Downloaded Character (Writable Character) Define Command ([ESC] &) Downloaded Character Erase Command ([ESC] ?)

6.10 Downloaded Character (Writable Character) define command [ESC] &

Function	Defines downloaded characters (writable characters) of the ANK code.
Format	[ESC] &nym [xd ₁ d ₂ d ₃ ...d _{yx}] _n ...[xd ₁ d ₂ d ₃ ...d _{yx}] _m
Term	ASCII: ESC & Hex: 1Bh 26h [Parameter] y: No. of bytes in the vertical direction 3 (03h) Fixed n: Setting start character code 32 ≤ n ≤ 255 (20h ≤ n ≤ FFh) m: Setting end character code 32 ≤ m ≤ 255 (20h ≤ m ≤ FFh) * n = m if there is only 1 character. d: Data to be defined 00h ≤ d₁, d₂, ... d_{yx} ≤ FFh x: No. of bytes in the horizontal direction Font Size 1 0 ≤ x ≤ 12 (00h ≤ x ≤ 0Ch) 0 ≤ x ≤ 9 (00h ≤ x ≤ 09h) Font Size 2 0 ≤ x ≤ 13 (00h ≤ x ≤ 0Dh) 0 ≤ x ≤ 10 (00h ≤ x ≤ 0Ah)
Default value	Internal ANK characters (no writable characters registered)
Explanation	(1) Multiple consecutive character codes can be defined in the single execution of the command. (2) d refers to not only data to be defined, but also a pattern of dots from the left to the x (horizontal) direction. If x is below the number of dots to form a character, the remaining dots on the right will be space filled. (3) At a moment when an error occurs with parameter y, n, m, or x due to a factor such as an unmatched condition, command processing will be aborted, and the data subsequent to x will be processed as normal data. (4) The defined downloaded characters are cleared in the event of the following commands: • Initialize Command ([ESC] @) • Downloaded Bit image Define Command ([GS] *) • Downloaded Character Erase Command ([ESC] ?)
Reference	[ESC] %, [ESC] ?

[Program example]

```

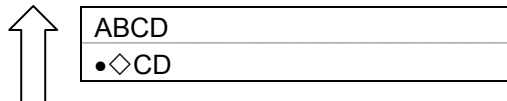
/* Data for registration of writable characters */
unsigned char gaiji1[3] = { 0x10, 0x00, 0x00 };
unsigned char gaiji2[15] = { 0x10, 0x00, 0x00, 0x28, 0x00, 0x00, 0x44, 0x00, 0x00,
0x28, 0x00, 0x00, 0x10, 0x00, 0x00 };

fprintf( stdprn, "\033&\003AB");          /* Registration of writable characters 41H,42H code
*/

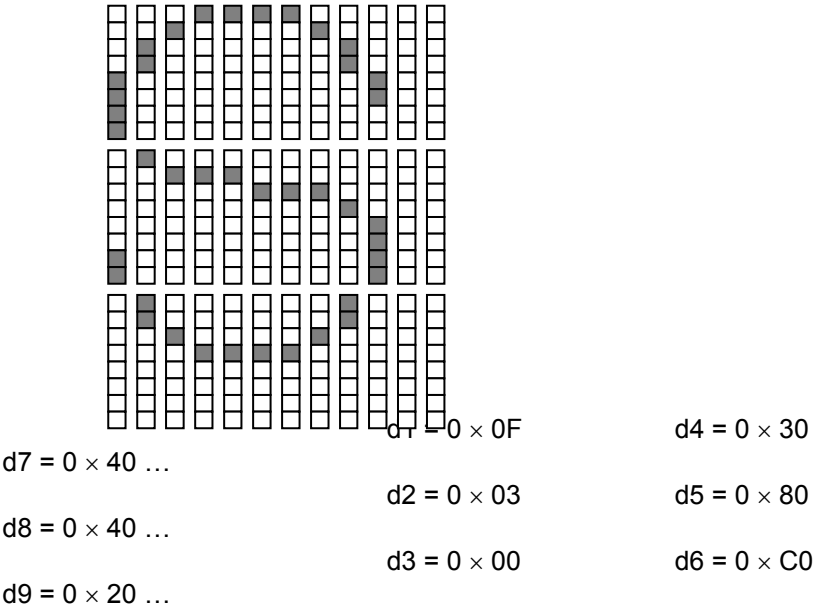
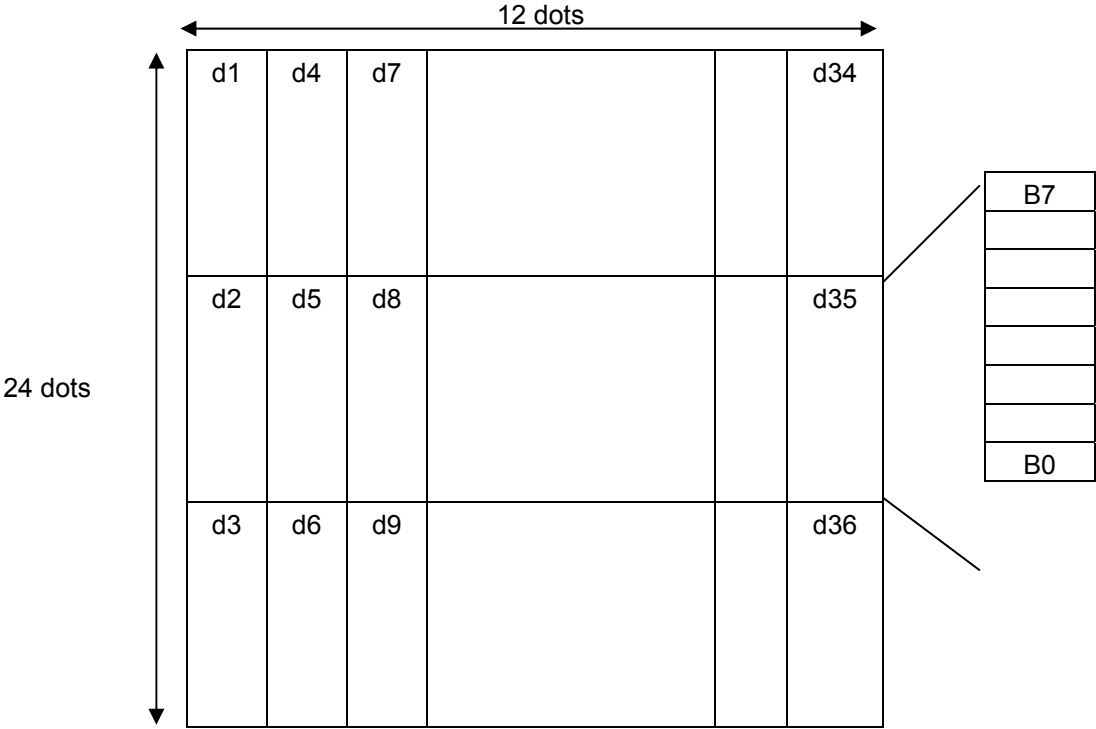
fputc( 1, stdprn );                        /* 1 dot in the horizontal direction */
for(i=0;i<3;i++)
    fputc( gaiji1[i], stdprn );
fputc( 5, stdprn );                        /* 5 dots in the horizontal direction */
for(i=0;i<15;i++)
    fputc( gaiji2[i], stdprn );
fprintf( stdprn, "\033%%");                /* Canceling writable characters */
fputc( 0, stdprn );
fprintf( stdprn, "ABCD\n");
fprintf( stdprn, "\033%%");                /* Setting writable characters */
fputc( 1, stdprn );
fprintf( stdprn, "ABCD\n");

```

[Print example]



[Specified character] Example of size1, font A



6.11 Bit image Mode Select command [ESC] *

Function Selects bit image mode (or prints image data).

Format [ESC] *mnLnH (d1d2...d_{((n1 + 256 × n2) × k)})

Term ASCII: ESC *
Hex: 1Bh 2Ah

[Parameter]

m: Mode

m = 0, 1, 32, 33 (m =

00h, 01h, 20h, 21h)

nL: No. of bytes in the horizontal direction (low order) $0 \leq nL \leq 255$ (00h ≤ nL ≤

FFh)

nH: No. of bytes in the horizontal direction (high order) $0 \leq nH \leq 3$ (00h ≤

nH ≤ 03h)

d: Image data

00h ≤ d₁, d₂, ... d_{yx} ≤ FFh

k: No. of bytes in the vertical direction

k = 1 if m = 0 or 1 (m = 00h k = 01h)

k = 3 if m =

32 or 33 (m = 20h, k = 21h)

Explanation

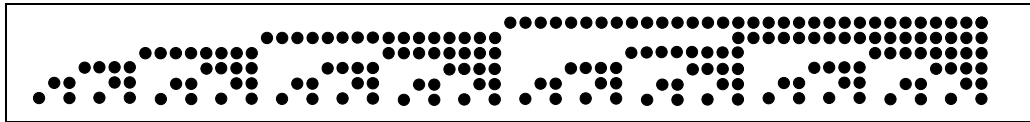
- (1) Image data m is specified by the number of dots defined by nL, nH and k.
- (2) The number of dots in the horizontal direction is determined by $nL + 256 \times nH$.
- (3) If the number of dots in the horizontal direction ($nL + 256 \times nH$) exceeds the value listed in the table below, printing will be performed while wrapping around, and consequently corrupted.

B-FP3D-GH/GS (203 dpi)	80 mm
------------------------	-------

- (4) d refers to image data. A bit to be printed is set to 1, and the one not to be printed is set to 0.
- (5) The table below shows mode selected by m.

m (mode)	No. of dots in the vertical direction	Dot density
0	8	Single density
1	8	Double density
32	24	Single density
33	24	Double density

- (6) Underlining, bold printing, reverse printing (white on black background), or inverted printing (upside-down) is not applicable to this image data.
- (7) A bit image in page mode is valid only when 0 (0-degree rotation) is specified by the Page Mode Print Direction Set Command ([ESC] T). If anything other than 0 degree is selected, the bit image will not be rotated.
- (8) At a moment when an error occurs with parameter m, nL, or nH due to a factor such as an unmatched condition, command processing will be aborted, and the subsequent data subsequent to x will be processed as normal data.
- (9) The most significant bit is printed on top, and the least significant at the bottom.



[Program example]

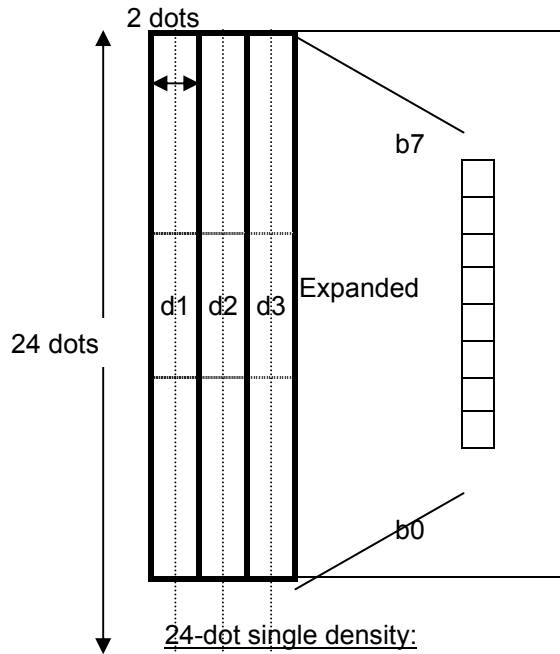
```
for(m=0;m<1;m++) {  
    fprintf( stdprn, "%033*%c%x3F%c", m, 0 );  
    for(i=1;i<=0x3F;i++) {  
        fputc( i, stdprn );  
    }  
    fputc( '\n', stdprn );  
}  
fprintf( stdprn, "%n%n");
```

[Print example]

Single density (m = 0)

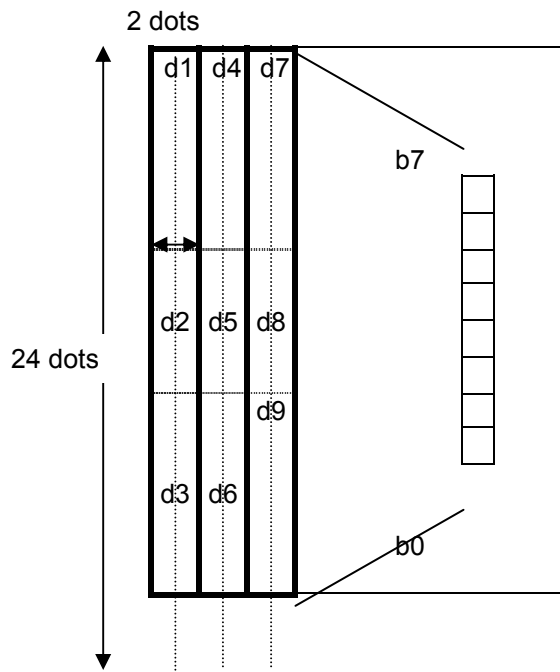
8-dot single density:

1-byte data is tripled in the vertical direction,
and doubled in the horizontal direction.



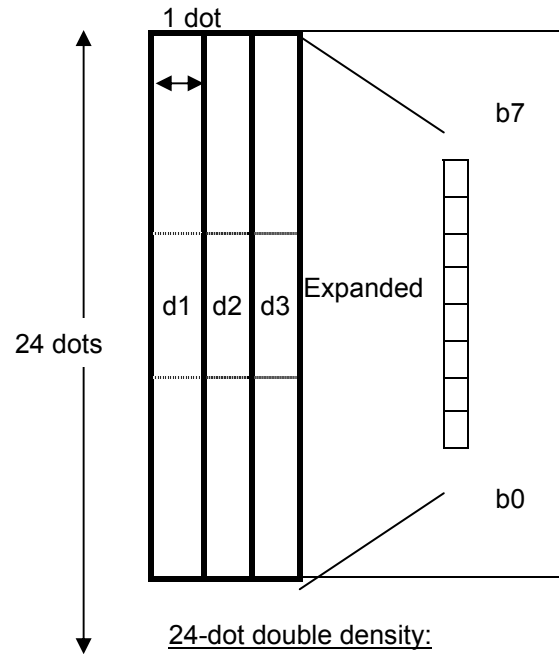
24-dot single density:

1-byte data is doubled in the horizontal
direction.



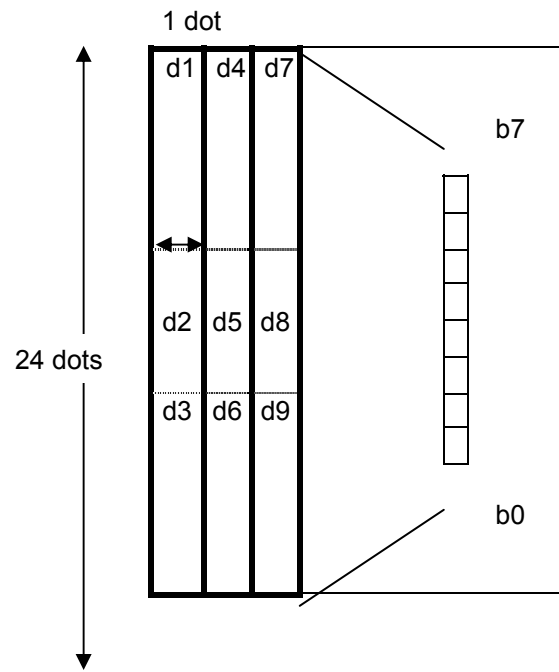
8-dot double density:

1-byte data is tripled in the vertical
direction.



24-dot double density:

1-byte data is edited in sequence.



6.12 Underlining Set command [ESC] -

Function	Sets or cancels underlining.
Format	[ESC] -n
Term	ASCII: ESC - Hex: 1Bh 2Dh [Parameter] n: Setting underlining $0 \leq n \leq 2$ (00h $\leq$ n $\leq$ 02h) or $48 \leq n \leq 50$ (30h $\leq$ n $\leq$ 32h) 0: Canceling underlining 1: Setting underlining (1-dot wide) 2: Setting underlining (2-dot wide)
Default value	n : 0
Explanation	(1) The character including the space on its right is underlined. However, a portion skipped by the Horizontal Tab Command ([HT]) is not underlined. (2) A character rotated 90 degrees clockwise or a character in reverse is not underlined. (3) If underlining is canceled with n = 0 or n = 48, subsequent data will not be underlined. However, the width of the underline specified most recently is maintained. Also, a 1-dot wide underline is selected by default. (4) The width of the underline is consistent regardless of a character size. (5) Underlining can also be set or canceled by the Print Mode All Set Command ([ESC] !), but the most recently processed command is applied. (6) There is no impact upon Chinese Characters.
Reference	Print Mode All Set Command ([ESC] !)

6.13 Downloaded Character Erase command [ESC] ?

Function	Erases downloaded characters in a specified code.		
Format	[ESC] ? n		
Term	ASCII: ESC ? Hex: 1Bh 3Fh [Parameter] n: Erasing downloaded characters FFh)		
		$32 \leq n \leq 255$ (20h ≤ n ≤	
Default value	n: 0		
Explanation	(1) n refers to a character code in which a defined pattern is erased. After downloaded characters are erased, the same pattern as the internal characters are printed. (2) If a specified character code is undefined, this command will be ignored.		
Reference	Downloaded Character (Writable Character) Define Command ([ESC] &) Downloaded Character (Writable Character) Set Command ([ESC] %)		

6.14 1/6 INCH Line Feed Length Set command [ESC] 2

Function	Sets a line feed length to 1/6 inch (34 dots). 0.133 Inch (3.33 mm) 0.167 Inch (4.21 mm)
Format	ASCII: ESC 2 Hex: 1Bh 32h
Explanation	(1) Line feed lengths can be specified in standard only.
Reference	Minimum Dots for Line Feed Length Set Command ([ESC] 3)

6.15 Minimum Dots for Line Feed Length Set command [ESC] 3

Function	Sets a line feed length to n dot. Specifies the minimum number of dots for line feed.	
Format	[ESC] 3 n	
Term	ASCII: ESC 3 Hex: 1Bh 33h [Parameter] n: Line feed length (n dots) $0 \leq n \leq 255$ (00h $\leq n \leq$ FFh)	
Default value	n: 27 (Line feed length of 27 dots)	
Explanation	(1) Line feed lengths can be independently specified in standard only. (2) Fractions are rounded up. (3) The minimum permissible value is 00h for a line feed length. However, a line feed length is only applicable when a value of 27 dots or over is specified. If a value less than 27 is specified, a line feed length will be changed to 27 dots.	
Reference	34-dot Line Feed Length Set Command ([ESC] 2)	

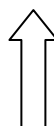
6.16 Initialize Command [ESC] @

Function	Clears data in the currently expanded image buffer, and initializes the settings.
Format	[ESC] @
Term	ASCII: ESC @ Hex: 1Bh 40h
Explanation	(1) The data and macro definition in the receive buffer are maintained. (2) Please refer to the next page for further information regarding the data to be initialized.

[Program example]

```
fprintf( stdprn, "\033!%c", 0x20 );      /* Double width */  
fprintf( stdprn, "11111\n");  
/*Other settings*/  
fprintf( stdprn, "\033@" );              /* Initializing printer */  
fprintf( stdprn, "22222\n");
```

[Print example]



1 1 1 1 1
2 2 2 2 2

 ← All settings are reset to default values.

[Data initialized by ESC@]

Command	Functions	Default value
HT	Horizontal tab	Every 8 characters (9th, 17th digit ...)
ESC SP	Spacing on the right of character	1 dot
ESC !	Character control	0 (No settings)
ESC %	Setting / canceling specified characters	0 (No writable characters specified)
ESC &	Definition of specified characters	Internal ANK characters (No writable characters registered)
ESC -	Setting / canceling underlining	0 (Canceling underline)
ESC 3	Minimum No. of dots for line feed length	27 (Line feed length of 27 dots)
ESC D	Horizontal tab position	
ESC E	Setting / canceling bold printing	0 (No bold printing)
ESC G	Setting / canceling double printing	0 (No double printing)
ESC T	Selecting print direction and starting point	0 (Left → right)
ESC V	Setting / canceling 90-degree rotation of character	0 (Canceling 90-degree rotation)
ESC W	Specifying print area	0,0,576,640 (B-FP3D-GH/GS)
ESC a	Specifying print position	0 (Left justified)
ESC c5	Enabling / disabling panel switch	0 (Enabled)
ESC {	Setting / canceling inverted printing	0 (Canceling inverted printing)
FS !	Specifying print modes for Chinese characters altogether	0 (No settings)
FS &	Specifying Chinese Character mode	Canceled
FS -	Setting / canceling underlining for Chinese characters	0 (No underlining)
FS .	Canceling Chinese character mode	0 (Chinese character mode canceled)
FS 2	Defining writable characters of Chinese characters	All space
FS C	Selecting Chinese character code system	0 (JIS code system)
FS S	Specifying spacing on both sides of Chinese character	Left: 0 dot Right: 2 dots
FS W	Setting / canceling quadrupled Chinese character (double height × double width)	0 (Canceled)
GS B	Setting / canceling reversed printing	0 (Reversed printing canceled)
GS w	Specifying width of barcode	3 (1 module 3 dots)
GS h	Specifying height of barcode	162 (20.25 mm)
GS H	Specifying position of HRI character of barcode	0 (No HRI printed)
GS f	Specifying font of HRI character of barcode	0 (HRI character = font A)
GS W	Specifying width of print area	576 dots (B-FP3D-GH/GS)
GS L	Specifying left margin	0 (No left margin)
GS !	Specifying character size	0 (Character size: × 1)

6.17 Horizontal tab positions set command [ESC] D

Function	Specifies the position of the horizontal tab.	
Format	[ESC] D [$n_1 n_2 n_3 \dots n_k$] [NUL]	
Term	ASCII: ESC D Hex: 1Bh 44h	
	[Parameter]	
	n: Horizontal tab position (No. of digits)	$1 \leq n \leq 255$ (01h $\leq n \leq$ FFh)
	k: Maximum number of horizontal tab positions	$0 \leq k \leq 32$ (00h $\leq k \leq$ 20h)
Default value	Every 8 characters from column. 1 (9, 17, 25, etc.) for normal print.	
Explanation	(1) The specified position of the horizontal tab is determined by [character width $\times$ n] from the left margin or the beginning of a line. "Character width" in this context refers to the width of a character including space on its right. Therefore, if a magnification is set to x 2 or more, the character width will be enlarged accordingly. (2) The position of the horizontal tab that has been specified is cleared. (3) With n = 8 for the position of the horizontal tab, the next print position will be shifted to the 9th digit by executing the Horizontal Tab Command ([HT]). (4) The maximum permissible number of horizontal tab positions is 32 (k = 32). If it is exceeded, the subsequent data will be processed as normal data. (5) Values are inputted to <n> for the position of the horizontal tab in ascending order, ending with (00H). If <n> is equal to or smaller than the previous <n>, the horizontal tab position setting will be aborted immediately, and the subsequent data will be processed as normal data. (6) All horizontal tab positions are cleared by [ESC] D [NUL]. (7) The specified horizontal tab position will remain unchanged even if the character width is changed after the horizontal tab position is specified.	

Example

01234567890123456789012345678901234[0A]
[09]H[09]H[09]H[09]H[0A]
[1B]D[0A][14][1E][00[09]H[09]H[09]H[09]H[0A]

[Program example]

```
fprintf( stdprn, "01234567890123456789012345678901234\n");  
fprintf( stdprn, "\033D%c%c%c", 10, 20, 30);  
fputc( 0, stdprn );  
for(i=0;i<4;i++) {  
    fputc( 't', stdprn );  
    fputc( 'H', stdprn );  
}  
fputc( '\n', stdprn );
```

[Print example]



01234567890123456789012345678901234															
H				H				H				H			
H				H				HH							

← Default position

Reference

Horizontal Tab Command ([HT])

6.18 Bold Printing Set command [ESC] E

Function Sets or cancels bold printing.

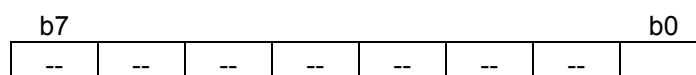
Format [ESC] En

Term ASCII: ESC E
Hex: 1Bh 45h

[Parameter]

n: Bold printing

$0 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh)



0: Canceling bold printing
1: Setting bold printing

Default value n: 0

- Explanation**
- (1) n is valid only in the least significant bit.
 - (2) In bold printing, one character is shifted 1 dot horizontally and superimposed on another. However, in the event of the Character Rotation Set Command ([ESC]V), bold printing is performed after the rotation of the character. Therefore the direction in which bold is applied is different.
 - (3) Bold printing can also be set or canceled by the Print Mode All Set Command ([ESC] G), but the most recently processed command is applied.
 - (4) The same print output is obtained as when the Double Printing Set Command ([ESC] G) is executed.

Reference Print Mode All Set Command ([ESC] !)

6.19 Double Printing Set command [ESC] G

Function Sets or cancels double printing.

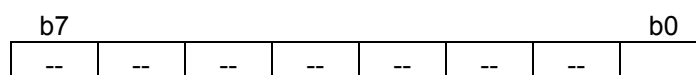
Format [ESC] G n

Term ASCII: ESC G
Hex: 1Bh 47h

[Parameter]

n: Double printing

$0 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh)



0: Canceling double printing
1: Setting double printing

Default value n: 0

Explanation

- (1) n is valid only in the least significant bit.
- (2) The same print output is obtained as when the Bold Printing Set Command ([ESC]E) is executed.

Reference Bold Printing Set Command ([ESC] E)

6.20 Feed Length Set command [ESC] J

Function	Prints data and feeds over the length of n dots.
----------	--

Format	[ESC] J n
--------	-----------

Term	ASCII: ESC J Hex: 1Bh 4Ah
------	------------------------------

[Parameter]

n: Feed length $0 \leq n \leq 255$ (00h $\leq n \leq$ FFh)

- | | | | |
|------------------------|--|------------------------|--------|
| Explanation | <p>(1) After this command is executed, the next print starting position is the beginning of a line.</p> <p>(2) There is no impact upon the line feed length specified by the 34-dot Line Feed Length Set Command ([ESC] 2) or Minimum Dots for Line Feed Length Set Command ([ESC] 3).</p> <p>(3) Fractions are rounded up, if any.</p> <p>(4) In standard mode, dots in the vertical direction (y) are used.</p> <p>(5) In page mode, the printer operation varies depending on the starting position, as follows:</p> <ul style="list-style-type: none">① If “top left” or “bottom right” is selected as the starting position by the Page Mode Print Direction Set Command ([ESC] T), the print position will be shifted in the paper feed direction (vertical direction of a character). In this case, dots in the vertical direction (y) are used.② If “top right” or “bottom left” is selected as the starting position by the Page Mode Print Direction Set Command ([ESC] T), the print position will be shifted in the direction perpendicular to the paper feed direction (vertical direction of a character). In this case, dots in the horizontal direction (x) are used. <p>(6) The value in the table below must be applied if the specified feed length exceeds it.</p> <table border="1"><tr><td>B-FP3D-GH/GS (203 dpi)</td><td>999 mm</td></tr></table> <p>(7) If the value set in the parameter of this command is smaller than the feed length of a character, printing will be performed with one character superimposed on another. Therefore, care must be taken to select the feed length.</p> | B-FP3D-GH/GS (203 dpi) | 999 mm |
| B-FP3D-GH/GS (203 dpi) | 999 mm | | |

6.21 Switching to Page Mode command [ESC] L

Function	Makes transition from standard mode to page mode
Format	[ESC] L
Term	ASCII: ESC L Hex: 1Bh 4Ch
Explanation	(1) This command will be valid only if it is included at the beginning of a line. (2) This command will be valid if it is included in page mode. (3) The printer returns to standard mode after printing ends with the Page Mode Print Command ([FF]), or with the Switching to Standard Mode Command ([ESC] S). (4) The print data is expanded within the print area specified by the Page Mode Print Area Set Command ([ESC] W). Also, the starting position is the position in the print area specified by the Page Mode Print Direction Set Command ([ESC] T). (5) The following commands have different values in page and standard modes. So, the values are switched to the ones for page mode. ① Spacing Set [ESC] [SP], [FS] S (6) The parameter settings made when the following commands are received in standard mode are applied after switching to page mode. ① [ESC] V Setting / canceling 90-degree rotation of character ② [ESC] a Justifying print position ③ [ESC] { Setting / canceling inverted printing ④ [GS] L Setting left margin ⑤ [GS] W Setting width of print area (7) The printer returns to standard mode by the Initialize Command ([ESC] @).
Reference	Page Mode Print Command ([FF]) Page Mode Cancel Command ([CAN]) Page Mode Collective Print Command ([ESC] [FF]) Switching to Standard Mode Command ([ESC] S) Page Mode Print Direction Set Command ([ESC] T) Page Mode Print Area Set Command ([ESC] W) Page Mode Relative Position in Vertical Direction Set Command ([GS] ¥)

6.22 Select International Character Set [ESC] R

Function Selects the international character set.

Format [ESC] Rn

Term ASCII: ESC R
Hex: 1Bh 52h

[Parameter]

n: Selecting international character set. $0 \leq n \leq 13$ (00h $\leq$ n $\leq$ 0Dh)

<i>n</i>	Character Set
0	USA
1	France
2	Germany
3	UK
4	Denmark I
5	Sweden
6	Italy
7	Spain
8	Japan
9	Norway
10	Denmark II
11	Spain II
12	Latin America
13	Republic of Korea

Default value n: 0

Explanation (1) Even if an international character set is switched between Japan and other countries by this command, there is no impact upon the Chinese character code system.

6.23 Switching to Standard Mode command [ESC] S

Function Makes transition from page mode to standard mode.

Format [ESC] S

Term ASCII: ESC S
Hex: 1Bh 53h

Explanation

- (1) This command will be valid only if it is executed in page mode.
- (2) The data expanded in page mode is erased.
- (3) After this command is executed, the next print starting position is the beginning of a line.
- (4) The following commands have different values in page and standard modes. So, the values are switched to the ones for standard mode.

① Spacing Set [ESC] [SP], [FS] S
- (5) Standard mode is selected at power-on, when resetting, or when the Initialize Command ([ESC] @) is executed.
- (6) The print area defined by the Page Mode Print Area Set Command ([ESC] W) is

initialized.

Reference

Page Mode Print Command ([FF])

Page Mode Collective Print Command ([ESC] [FF])

Switching to Page Mode Command ([ESC] L)

6.24 Page Mode Print Direction Set command [ESC] T

Function Selects a print direction and print starting point in page mode.

Format [ESC] T n

Term ASCII: ESC T
Hex: 1Bh 54h

[Parameter]

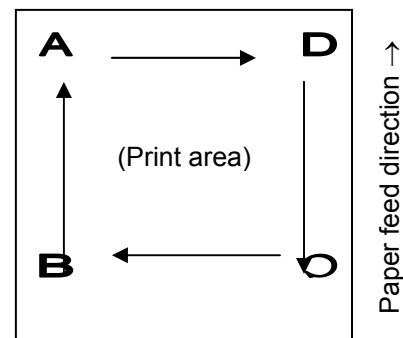
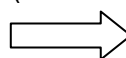
n: Selecting print direction and starting point

$0 \leq n \leq 3$ (00h $\leq$ n $\leq$ 03h)

$48 \leq n \leq 51$ (30h $\leq$ n $\leq$ 33h)

N	Print direction	
0, 48	Left → Right	Top left (A in the figure on the right)
1, 49	Bottom → Top	Bottom left (B in the figure on the right)
2, 50	Right → Left	Bottom right (C in the figure on the right)
3, 51	Top → Bottom	Top right (D in the figure on the right)

(Reference)



Default value n: 0

- Explanation**
- (1) If this command is received while the printer is in standard mode, the defined parameter setting will be applied after switching to page mode.
 - (2) The position where characters are expanded is the starting position in the print area defined by the Page Mode Print Area Set Command ([ESC] W).

Reference

Switching to Standard Mode Command ([ESC] S)
Switching to Page Mode Command ([ESC] L)
Page Mode Print Area Set Command ([ESC] W)
Relative Position Set Command ([ESC] ¥)
Page Mode Relative Position in Vertical Direction Set Command ([GS] ¥)

6.25 Character Rotation Set command [ESC] V

Function	Sets or cancels the 90-degree rotation (clockwise) of a character.
----------	--

Format	[ESC] V n
--------	-----------

Term	ASCII: ESC V Hex: 1Bh 56h
------	------------------------------

[Parameter]

n: Selecting 90-degree rotation (clockwise) $0 \leq n \leq 1$ (00h $\leq n \leq$ 01h)
or
 $48 \leq n \leq 49$ (30h $\leq n \leq$ 31h)

N	
0, 48	Canceling 90-degree of rotation (clockwise)
1, 49	Setting 90-degree of rotation (clockwise)

Default value	n: 0
---------------	------

Explanation	(1) Even if underlining is set, a character rotated 90 degrees clockwise will not be underlined. (2) If 90-degree of rotation (clockwise) is set, a relationship between double width and double height is the reverse of the one when 90-degree rotation is canceled. (3) The command setting does not have any impacts upon page mode. (4) If this command is received while the printer in page mode, the defined parameter setting will be applied after switching to standard mode.
-------------	--

Reference	Print Mode All Set Command ([ESC] !) Underlining Set Command ([ESC] -)
-----------	---

6.26 Page Mode Print Area Set command [ESC] W

Function	Specifies a position and size of the print area in page mode.		
Format	[ESC] W xL xH yL yH dxL dxH dyL dyH		
Term	ASCII: ESC W Hex: 1Bh 57h [Parameter] xL : Starting point in horizontal direction (low order) $0 \leq xL \leq 255$ (00h $\leq$ xL $\leq$ FFh) xH : Starting point in horizontal direction (high order) $0 \leq xH \leq 255$ (00h $\leq$ xH $\leq$ FFh) yL : Starting point in vertical direction (low order) $0 \leq yL \leq 255$ (00h $\leq$ yL $\leq$ FFh) yH : Starting point in vertical direction (high order) $0 \leq yH \leq 255$ (00h $\leq$ yH $\leq$ FFh) dxL : Length in horizontal direction (low order) $0 \leq dxL \leq 255$ (00h $\leq$ dxL $\leq$ FFh) dxH : Length in horizontal direction (high order) $0 \leq dxH \leq 255$ (00h $\leq$ dxH $\leq$ FFh) dyL : Length in vertical direction (low order) $0 \leq dyL \leq 255$ (00h $\leq$ dyL $\leq$ FFh) dyH : Length in vertical direction (high order) $0 \leq dyH \leq 255$ (00h $\leq$ dyH $\leq$ FFh) * Starting point in horizontal direction = xL + xH $\times$ 256 dots Starting point in vertical direction = yL + yH $\times$ 256 dots Length in horizontal direction = dxL + dxH $\times$ 256 dots Length in vertical direction = dyL + dyH $\times$ 256 dots		
Default value	B-FP3D-GH/GS: xL=xH=yL=yH=0, dxL=40h dxH=02h dyL=80h dyH=02h		
Explanation	(1) If this command is received while the printer in standard mode, the defined parameter setting will be applied after switching to page mode. (2) If the starting point in the horizontal or vertical direction is out of the print area, command processing will be aborted, and the subsequent data will be processed as normal data. (3) If the length in the horizontal or vertical direction is 0, command processing will be aborted, and the subsequent data will be processed as normal data. (4) The position where characters are expanded is the starting position in the print area defined by the Page Mode Print Direction Set Command ([ESC] T).		

- (5) At a moment when an error occurs with parameter xL to dyH due to a factor such as an unmatched condition, command processing will be aborted, and the data subsequent to dyH will be processed as normal data.
- (6) If the following commands, which are used commonly among the B-FP series, are received, a higher priority will be given to them. Therefore, the same range as these commands cannot be specified.

[ESC] WS [LF] [NUL]
[ESC] WR [LF] [NUL]
[ESC] WX [LF] [NUL]

(Example)

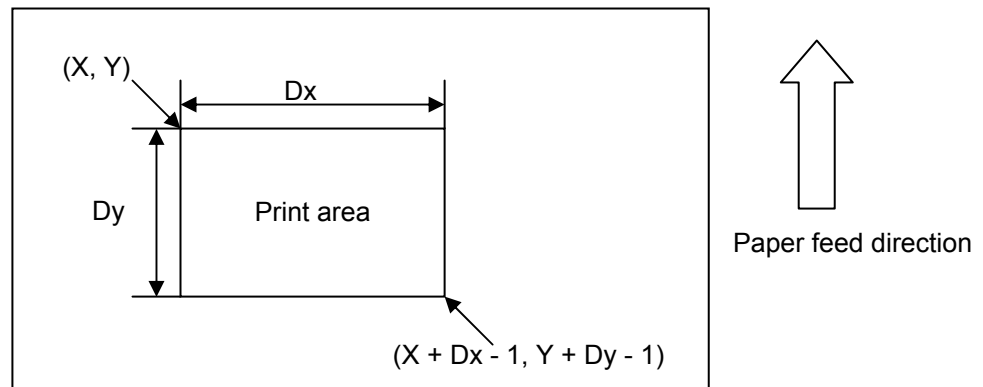
[ESC] W	<u>xL</u>	<u>xH</u>	<u>yL</u>	<u>yH</u>	<u>dxL</u>	<u>dxH</u>	<u>dyL</u>	<u>dyH</u>
	↓	↓	↓	↓	↓	↓	↓	↓
	S	10	0	0	160	1	160	1

The above values are specified. If inputted in hex format, they are as follows:

[1BH][57H][53H][0AH][00H][00H][A0H][01H][A0H][01H]

However the underlined portion is considered as [ESC] WS [LF] [NUL]. Therefore, a priority is given to the Status Request Command rather than the Page Mode Print Area Set Command.

- (7) The figure below illustrates a print area with the starting point in the horizontal direction X, starting point in the vertical direction Y, length in the horizontal direction Dx, and length in the vertical direction Dy.



6.27 Justification command [ESC] a

Function	Justifies print data.		
Format	[ESC] a n		
Term	ASCII: ESC a Hex: 1Bh 61h		
	[Parameter]		
	n: Justifying print position	0 ≤ n ≤ 2 (00h ≤ n ≤ 02h) or 48 ≤ n ≤ 50 (30h ≤ n ≤ 32h)	
	0: Left-justifying		
	1: Centering		
	2: Right-justifying		
Default value	n: 0		
Explanation	(1) This command will be executed only if it is received at the beginning of a line. (2) If this command is received while the printer in page mode, the defined parameter setting will be applied after switching to standard mode. (3) Justification is performed within the specified print area. (4) Justification also applies to a portion skipped by the Horizontal Tab Command ([HT]), Absolute Position Set Command ([ESC] \$), or Relative Position Set Command ([ESC] ¥).		

6.28 Panel key (Feed key) set command [ESC] C5

Function	Enables or disables the panel key (FEED key).
----------	---

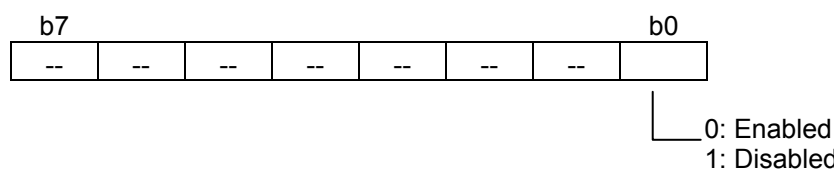
Format	[ESC] c5n
--------	-----------

Term	ASCII: ESC c 5 Hex: 1Bh 63h 35h
------	------------------------------------

[Parameter]

n: Selecting / deselecting printer

$0 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh)



Default value	n: 0
---------------	------

Explanation	(1) n is valid only in the least significant bit. (2) The [FEED] key will be disabled if the parameter is set to "Disabled." (3) While waiting for the [FEED] key entry when a macro is executed, the [FEED] key is enabled regardless of the parameter setting of this command. However, paper feed is not performed.
-------------	--

6.29 Paper Feed Length Set command [ESC] d

Function	Prints data and feeds paper over n lines. However, the specified number of lines that paper is fed over is not stored.
----------	--

Format	[ESC] d
--------	---------

Term	ASCII: ESC d Hex: 1Bh 64h
------	------------------------------

[Parameter]

n: No. of lines that paper is fed over $0 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh)

Explanation	(1) After this command is executed, the next print starting position is the beginning of a line.
-------------	--

(2) The value in the table below must be applied if $n \times$ line feed length exceeds it.

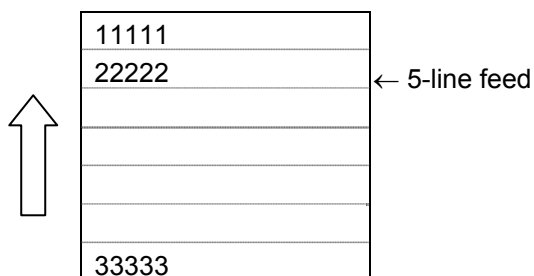
B-FP3D-GH/GS (203 dpi)	999 mm
------------------------	--------

(3) The line feed length is 27 dots per line.

[Program example]

```
fprintf( stdprn, "11111¥n");  
fprintf( stdprn, "22222");  
fprintf( stdprn, "¥033d¥x05");          /* 5-line feed */  
fprintf( stdprn, "33333¥n");
```

[Print example]



6.30 SELECTS CHARACTER code TABLE [ESC] t

Function	Selects a character code table		
Format	[ESC] t n		
Term	ASCII: ESC t Hex: 1Bh 74h		
	[Parameter]		
	n: Selecting code page No.	0 ≤ n ≤ 13 (00h ≤ n ≤ 0Dh)	
	0: PC437 (USA: Standard Europe)		
	1: Not Used		
	2: PC850 (Multilingual)		
	3: PC860 (Portuguese)		
	4: PC863 (Canadian-French)		
	5: PC865 (Nordic)		
	6: PC852		
	7: PC866		
	8: PC857		
	9: PC1252		
	10: PC858		
	11: PC1253		
	12: PC1254		
	13: PC1256		
Default value	n: 0		
Explanation	(1) The character sets cannot be used together on the same line.		

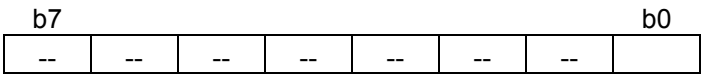
6.31 Inverted Printing Set command [ESC] {

Function Sets or cancels inverted printing (upside-down).

Format [ESC] {n

Term ASCII: ESC {
Hex: 1Bh 7Bh

[Parameter]
n: Setting/canceling inverted printing 0 ≤ n ≤ 255 (00h ≤ n ≤ FFh)



0: Canceling inverted printing
1: Setting inverted printing

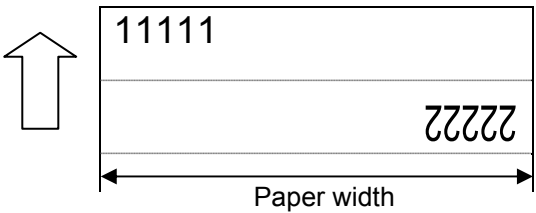
Default value n: 0

- Explanation
- (1) n is valid only in the least significant bit.
 - (2) This command is executed only if it is received at the beginning of a line.
 - (3) Inverted printing means that a line is inverted 180 degrees and printed.
 - (4) Inverted printing is performed in the direction opposed to normal printing.
 - (5) If this command is received while the printer is in page mode, the defined parameter setting will be applied after switching to standard mode.

[Program example]

```
fprintf( stdprn, "\033{");          /* Canceling inverted printing */
fputc( 0, stdprn );
fprintf( stdprn, "11111\n");
fprintf( stdprn, "\033{");          /* Setting inverted printing */
fputc( 1, stdprn );
fprintf( stdprn, "22222\n");
```

[Printer example]



6.32 Absolute Position Set command [ESC] \$

Function	Specifies an absolute position.	
Format	[ESC] \$ nL nH	
Term	ASCII: ESC \$ Hex: 1Bh 24h	
	[Parameter]	
	n: No. of dots to specify absolute position (low order) ≤ nL ≤ FFh)	0 ≤ nL ≤ 255 (00h
	n: No. of dots to specify absolute position (high order) ≤ nH ≤ FFh)	0 ≤ nH ≤ 255 (00h
Explanation	(1) An absolute position of the subsequent print starting position is specified with reference to the left margin. The next print starting position is located (nL + nH × 256) / dots away from the left margin. (2) An absolute position out of the print area is ignored. (3) Dots in the horizontal direction (x) are applied in standard mode. (4) The printer operation varies in page mode depending on the starting position as follows: ① If “top left” or “bottom right” is selected as the starting point defined by the Page Mode Print Direction Set Command ([ESC] T), an absolute position in the direction perpendicular to the paper feed direction (horizontal direction of a character) is specified. ② If “top right” or “bottom left” is selected as the starting point defined by the Page Mode Print Direction Set Command ([ESC] T), an absolute position in the paper feed direction (horizontal direction of a character) is specified.	
Reference	Relative Position Set Command ([ESC] ¥) Page Mode Absolute Position in Vertical Direction Set Command ([GS] \$) Page Mode Relative Position in Vertical Direction Set Command ([GS] ¥)	

6.33 Relative Position Set command [ESC] ¥

Function	Specifies a relative position.		
Format	[ESC] ¥ nL nH		
Term	ASCII: ESC ¥ Hex: 1Bh 5Ch [Parameter] n: No. of dots to specify relative position (low order) $0 \leq nL \leq 255$ (00h $\leq$ nL $\leq$ FFh) n: No. of dots to specify relative position (high order) $0 \leq nH \leq 255$ (00h $\leq$ nH $\leq$ FFh)		
Default value	Not defined.		
Explanation	(1) A relative position of the subsequent print starting position is specified with reference to a current position. The next print starting position is located (nL + nH × 256) away from the current position. (2) A relative position out of the print area is ignored. (3) If the right direction of the current position is specified, a positive number must be defined. If the left direction of the current position is specified, a negative number must be defined. (4) A negative number is represented by a complement number of 65535. Ex.) To specify N dots in the left (negative) direction $nL + nH \times 256 = 65536 - N.$ (5) Dots in the horizontal direction (x) are applied in standard mode. (6) The printer operation varies in page mode depending on the starting position as follows: ① If “top left” or “bottom right” is selected as the starting point in the Page Mode Print Direction Set Command ([ESC] T), a relative position in the direction perpendicular to the paper feed direction (horizontal direction of a character) is specified. ② If “top right” or “bottom left” is selected as the starting point in the Page Mode Print Direction Set Command ([ESC] T), a relative position in the paper feed direction (horizontal direction of a character) is specified.		
Reference	Absolute Position Set Command ([ESC] \$)		

6.34 KANJI character Print Mode All Set command [FS] !

Function Specifies printing mode for Kanji characters altogether.

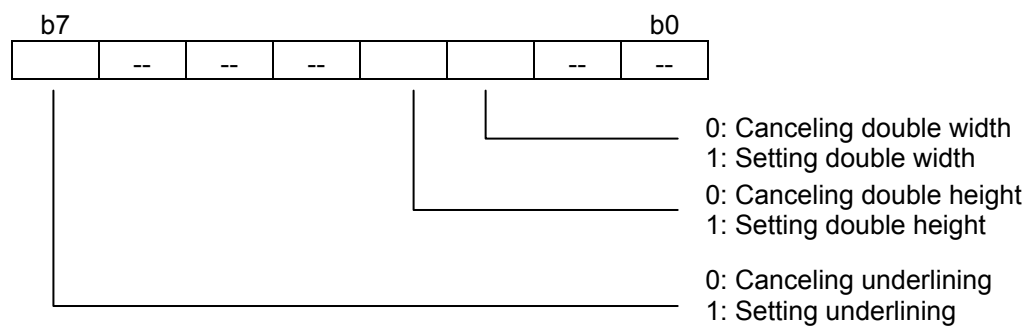
Format [FS] !

Term ASCII: FS!
Hex: 1Ch 21h

[Parameter]

n: Specifying character control of Kanji character
 $n \leq FFh$)

$0 \leq n \leq 255$ (00h $\leq$



Default value n: 0

6.35 Kanji Character Mode Set command [FS] &

Function	Prints Kanji characters.
Format	[FS] &
Term	ASCII: FS& Hex: 1Ch 26h
Explanation	(1) If this command is received with shift JIS selected for a Kanji character code system, the defined parameter setting will be applied when the Kanji character code system is changed to JIS. There is no impact upon printing. (2) Kanji character mode is canceled by default. (3) A Kanji character code is processed from the 1st to the 2nd byte. (4) The parameter setting applies to the Kanji language settings with Kanji language implemented for the QM model (destined overseas), or to the Korean language setting with Korean language implemented.
Reference	Kanji Character Mode Cancel Command ([FS] .) Kanji Character Code System Set Command ([FS] C)

6.36 Kanji Character Underlining Set command [FS] -

Function	Sets or cancels underlining for Kanji characters.		
Format	[FS]-n		
Term	ASCII: FS- Hex: 1Ch 2Dh		
	[Parameter]		
	n: Setting underlining for Chinese character	$0 \leq n \leq 2$ (00h $\leq n \leq$ 02h)	
		or	
		$48 \leq n \leq$	
	50 (30h $\leq n \leq$ 32h)		
	0: Canceling underlining		
	1: Setting underlining (1-dot wide)		
	2: Setting underlining (2-dot wide)		
Default value	n: 0		
Explanation	(1) Underline Kanji characters can also be set or canceled by the Kanji Character Print Mode All Set Command ([FS] !), but the most recently processed command is applied. (2) The parameter setting applies to the Kanji language settings with Kanji language implemented for the QM model (destined overseas), or to the Korean language setting with Korean language implemented.		
Reference	Kanji Character Print Mode All Set Command ([FS] !)		

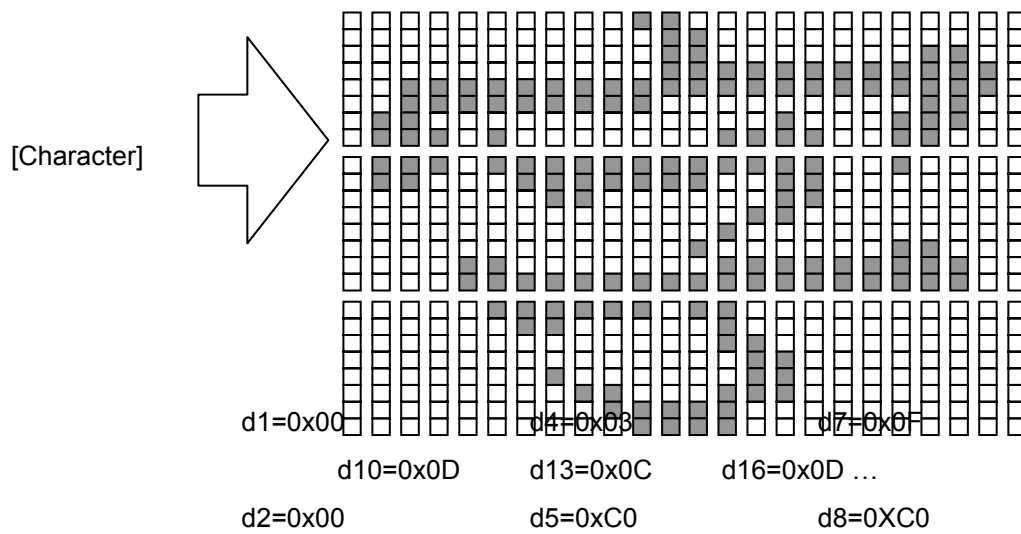
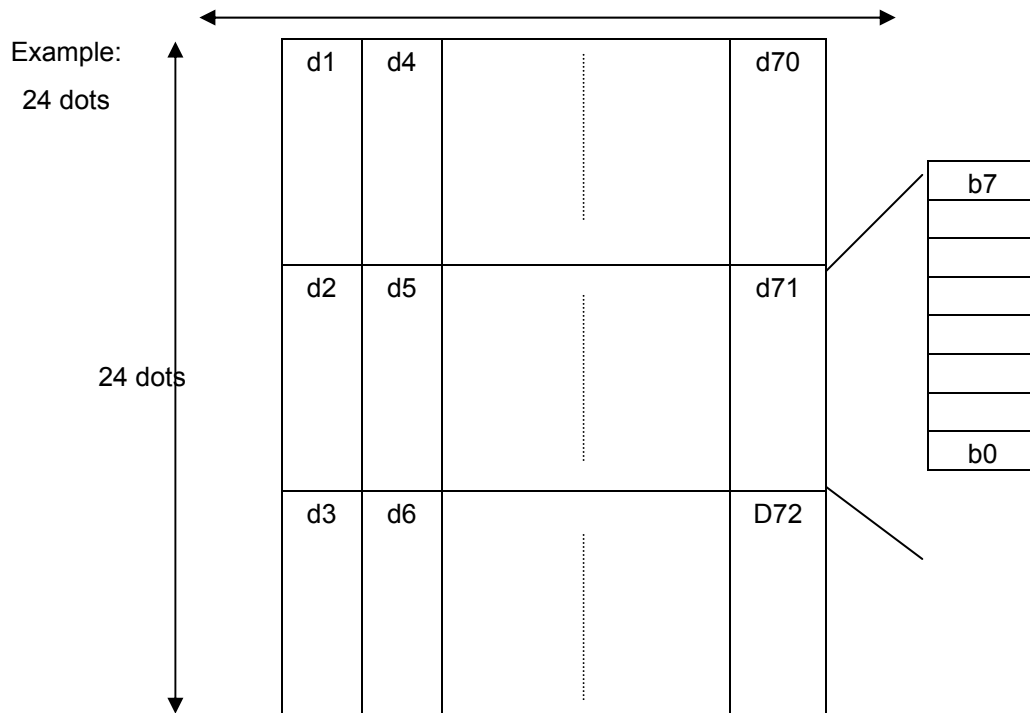
6.37 Kanji Character Mode Cancel command [FS] .

Function	Cancels Kanji character mode.
Format	[FS] .
Term	ASCII: FS. Hex: 1Ch 2Eh
Explanation	(1) If this command is received with shift JIS selected for a Kanji character code system, the defined parameter setting will be applied when the Kanji character code system is changed to JIS. There is no impact upon printing. (2) Kanji character mode is canceled by default.
Reference	Kanji Character Mode Set Command ([FS] &) Kanji Character Code System Set Command ([FS] C)

6.38 Writable Character of Chinese Character Define command [FS] 2

Function	Specifies writable characters of Chinese characters.
Format	[FS] 2 a1 a2 [d ₁ d ₂ d ₃ ...d _k]
Term	ASCII: FS2 Hex: 1Ch 32h [Parameter] a1: Writable character code (high order) a2: Writable character code (low order) JIS code: a1 = 77h, 21h ≤ a2 ≤ 7Eh Shift JIS: a1 = ECh, 40h ≤ a2 ≤ 9Eh (except for 7Fh) EUC code: a1 = F7h, A1h ≤ a2 ≤ FEh Chinese/Korean: a1 = FFh, 40h ≤ a2 ≤ 7Eh, 80h ≤ a2 ≤ FCh d: Data to be defined 0 ≤ d ≤ 255 Data consists of 72 bytes (k = 72) (3 bytes (high) × 24 dots)
Default value	All space
Explanation	(1) a1 and a2 refer to a Kanji character code to define writable characters. a1 refers to the 1st byte, and a2 the 2nd byte. (2) d refers to data to be defined. A bit corresponding to a dot to be printed is 1, and the one corresponding to a dot not to be printed is 0. (3) At a moment when an error occurs with parameter a1 or a2 due to a factor such as an unmatched condition, command processing will be aborted, and the data subsequent to a2 will be processed as normal data. (4) The defined writable characters apply until resetting or until the printer is turned off.
Reference	Kanji Character Code System Set Command ([FS] C)

[Defined character] Example of font A



6.39 Kanji Character Code System Set command [FS] C

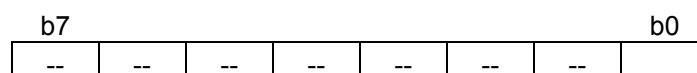
Function Selects a Kanji character code system.

Format [FS] C n

Term ASCII: FS C
Hex: 1Ch 43h

[Parameter]

n: Selecting Chinese character code system n = 0, 1 (n = 00h, 01h)



└─ 0: JIS code system
1: Shift JIS code system

Default value n: 0

- Explanation**
- (1) For JIS code, valid Kanji character codes are 21 to 7Eh for both the 1st and 2nd bytes.
 - (2) For shift JIS code, valid Kanji character codes are as follows:
 1st byte: 81h to 9Fh, or E0h to EFh
 2nd byte: 40h to 7Eh, or 80h to FCh
 - (3) There is no impact upon the setting with Chinese or Korean language implemented.

[Program example]

```

fprintf( stdprn, "\034C\x01");          /* SHIFT-JIS selected */
fprintf( stdprn, "\x44\x7b\n");
fprintf( stdprn, "\034C%c",0x00);        /* JIS selected */
fprintf( stdprn, "\034&");              /* Beginning of Kanji character */
fprintf( stdprn, "\x44\x7b\n");
fprintf( stdprn, "\034.");               /* End of Kanji character */
fprintf( stdprn, "\x44\x7b\n");

```

[Print example]



D{
訂
D{

[Example]

- ① <SJIS>123ABCあいう<ANK>アウ
[1C][43][01][1C][26][31][32][33][41][42][43][82][A0][82][A2][82][A4][1C][2E][B1][B2][B3]
- ② <ANK>123ABC<JIS>あいう<ANK>アイウ
[1C][2E][31][32][33][41][42][43][1C][43][00][1C][26][24][22][24][24][24][26][1C][2E][B1][B2][B3]
- ③ <ANK>123ABC<SJIS>あいう<ANK>アイウ
[1C][2E][31][32][33][41][42][43][1C][43][01][1C][26][82][A0][82][A2][82][A4][1C][2E][B1][B2][B3]
- ④ <SJIS>123ABCあいうアウ
[1C][43][01][1C][26][31][32][33][41][42][43][82][A0][82][A2][82][A4][B1][B2][B3]
- ⑤ <ANK>123ABC<SJIS>あいうアウ
[1C][2E][31][32][33][41][42][43][1C][43][01][1C][26][82][A0][82][A2][82][A4][B1][B2][B3]
- ⑥ <ANK>123ABC<JIS>あいうアウ
[1C][2E][31][32][33][41][42][43][1C][43][00][1C][26][24][22][24][24][24][26][B1][B2][B3]

Reference

Chapter 12 Character Code Table

Chapter 13 Chinese Character Code Table

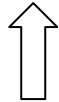
6.40 Chinese Character Spacing Set command [FS] S

Function	Specifies spacing (or the number of dots) on both sides of a Chinese character.		
Format	[FS] S n1 n2		
Term	ASCII: FS S Hex: 1Ch 53h [Parameter] n1: Spacing on the left of Kanji character 0 n2: Spacing on the right of Kanji character $0 \leq n2 \leq 32$ (00h $\leq$ n2 $\leq$ 20h)		
Default value	n1: 0, n2: 0		
Explanation	(1) Spacing defined this command is the one when a character size is standard. If a character with a magnification of x2 or over in the horizontal direction is expanded, spacing will be determined by the specified spacing $\times$ magnification in the horizontal direction. (2) The maximum permissible spacing on the right of a character is 32 dots. If a specified value exceeds 32, the command is ignored (3) Spacing can be independently specified in page and standard modes. (4) If a double-width character is specified, spacing will also be doubled. (5) For the B-FP series, spacing on the left of a Kanji character is added to spacing on its right. Therefore, there is no space found on the left of the Kanji character in the print output.		

[Program example]

```
fprintf( stdprn, "\034S%c\x02",0x00);      /* 2 dots on the right of character */  
fprintf( stdprn, "\x82\x50\x82\x50\x82\x50\x82\x50\x82\x50\n");  
fprintf( stdprn, "\034S%c\x1c",0x00);      /* 28 dots on the right of character */  
fprintf( stdprn, "\x82\x51\x82\x51\x82\x51\x82\x51\x82\x51\n");
```

[Print example]



1111
2 2 2 2

6.41 Kanji Character Magnification Set command [FS] W

Function Sets or cancels the quadrupling of Kanji character (double height × double width).

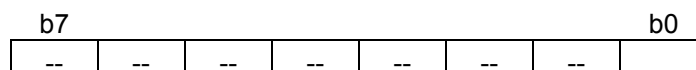
Format [FS] Wn

Term ASCII: FS W
Hex: 1Ch 57h

[Parameter]

n: Setting quadrupling of Kanji character (double height × double width)

$0 \leq n \leq 255$ (00h ≤ n ≤ FFh)



└ 0: Canceling quadrupling
1: Setting quadrupling

Default value n: 0

Explanation

- (1) n is valid only in the least significant bit.
- (2) If the quadrupling of a Kanji character (double height × double width) is canceled by this command, the subsequent Kanji character is printed in standard size (x 1).
- (3) A Kanji character size can also be specified by the Character Size Set Command ([GS] !) or Kanji Character Print Mode All Set Command ([FS] !), but the most recently processed command is applied.

Reference Kanji Character Print Mode All Set Command ([FS] !)
Character Size Set Command ([GS] !)

6.42 page mode Absolute Position Set command [GS] \$

Function	Specifies an absolute position in the vertical direction in page mode.
Format	[GS] \$nLnH
Term	ASCII: GS \$ Hex: 1Dh 24h [Parameter] nL: Specifying absolute position in the vertical direction in page mode (low order) $0 \leq nL \leq 255$ (00h $\leq$ nL $\leq$ FFh) nH: Specifying absolute position in the vertical direction in page mode (high order) $0 \leq nH \leq 255$ (00h $\leq$ nH $\leq$ FFh)
Default value	
Explanation	(1) An absolute position of the data expansion starting position in the vertical direction in page mode is specified with reference to the starting position. The next data expansion starting position is shifted ($nL + nH \times 256$) dots away from the current position in the vertical direction. (2) This command is ignored except when page mode is selected. (3) An absolute position in the vertical direction out of the specified print area is ignored. (4) A starting position, which is a reference to an absolute position, is specified in the Page Mode Print Direction Set Command ([ESC] T).
Reference	Absolute Position Set Command ([ESC] \$) Page Mode Print Direction Set Command ([ESC] T) Page Mode Print Area Set Command ([ESC] W) Relative Position Set Command ([ESC] \) Page Mode Relative Position in Vertical Direction Set Command ([GS] \)

6.43 Reverse character set command [GS] B

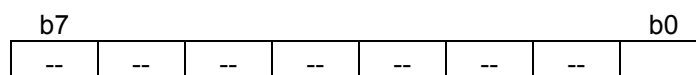
Function	Sets or cancels reverse characters (white text on black)
----------	--

Format	[FS] Bn
--------	---------

Term	ASCII: GS B Hex: 1Dh 42h
------	-----------------------------

[Parameter]

n: Setting reverse characters $0 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh)



└ 0: Canceling reverse character
1: Setting reverse character

Default value	n: 0
---------------	------

- | | |
|-------------|---|
| Explanation | <ul style="list-style-type: none">(1) n is valid only in the least significant bit.(2) Reversed printing applies to the internal and downloaded characters.(3) Reversed printing also applies to spacing on the right of a character defined by the Spacing Set Command ([ESC] [SP]).(4) There is no impact upon bit images, downloaded bit images, barcodes, or HRI characters, or a portion skipped by the Horizontal Tab Command ([HT]) or Relative Position Set Command ([ESC] \).(5) There is no impact upon spacing between lines.(6) A higher priority is given to reversed printing than underlining. Therefore, even if underlining is specified, reverse characters will not be underlined. However, the underlining settings will remain unchanged. |
|-------------|---|

6.44 ID transmission command [GS] I

Function Transmits the following ID specified by n.

Format [GS] In

Term ASCII: GS I
Hex: 1Dh 49h

[Parameter]

n: Selecting ID to transmit $1 \leq n \leq 3$ (01h $\leq$ n $\leq$ 03h)
or
 $49 \leq n \leq 51$ (31h $\leq$ n $\leq$ 33h)

n	ID type	ID to transmit
1	Model ID	0DH (13)
2	Type ID	See (2) below for further information.
3	ROM version ID	0x00
4	Logo Definition	See (3) below for further information

Default value n:

Explanation

- (1) This command is a batch mode command. The response is transmitted after all prior in the receive buffer has been processed. There may be a time lag between the printer receiving this command and transmitting the response.
- (2) Type ID is fixed at 00H for this printer.

Type ID (n=2)

bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined
	On	01	1	Undefined
1	Off	00	0	No knife installed
	On	02	2	knife installed
2	-	-	-	Undefined
3	-	-	-	Undefined
4	Off	00	0	Not used. Fixed to off.
5	-	-	-	Undefined
6	-	-	-	Undefined
7	Off	00	0	Not used. Fixed to off.

(3)

Type ID (n=4)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	No logo definition loaded by application.
	On	01	1	Logo loaded by application.
1	-	-	-	Undefined
2	-	-	-	Undefined

3	-	-	-	Undefined
4	Off	00	0	Not used. Fixed to off.
5	-	-	-	Undefined
6	-	-	-	Undefined
7	Off	00	0	Not used. Fixed to off.

6.45 Page Mode relative Position in Vertical Direction Set command [GS] ¥

Function	Specifies a relative position in the vertical direction in page mode.
Format	[GS] \nLnH
Term	ASCII: GS \ Hex: 1Dh 5Ch [Parameter] nL: Specifying relative position in the vertical direction in page mode (low order) $0 \leq nL \leq 255$ (00h $\leq$ nL $\leq$ FFh) nH: Specifying relative position in the vertical direction in page mode (high order) $0 \leq nH \leq 255$ (00h $\leq$ nH $\leq$ FFh)
Default value	
Explanation	(1) A relative position of the data expansion starting position in the vertical direction in page mode is specified with reference to a current position. The next data expansion starting position is shifted (nL + nH $\times$ 256) dots away from the current position in the vertical direction. (2) This command is ignored except when page mode is selected. (3) In JIS standard characters, “¥” is equivalent to “\”. (4) A relative position in the vertical direction out of the specified print area is ignored. (5) A positive number must be specified for the downward direction, and a negative number for the upward direction. (6) The number of dots is determined by $nH \times 256 + nL$. (7) N dots in the upward (negative) direction are specified by a complement number of N. - N dots = 65536 - N
Reference	Absolute Position Set Command ([ESC] \$) Page Mode Print Direction Set Command ([ESC] T) Page Mode Print Area Set Command ([ESC] W) Relative Position Set Command ([ESC] \) Page Mode Absolute Position in Vertical Direction Set Command ([GS] \$)

6.46 Barcode print command [GS] K

Function Selects a barcode system and prints barcodes.

Format ① [GS] k m [d₁d₂d₃...d_k] [NUL]

② [GS] k m n [d₁d₂d₃...d_n]

Term ASCII: GS k
Hex: 1Dh 6Bh

[Parameter]

m: Barcode type

①: $0 \leq m \leq 6, 10$ (00h $\leq m \leq$ 06h), 0Ah

The ranges of k and d vary depending on the selected barcode system.

②: $65 \leq m \leq 73, 75$ (41h $\leq m \leq$ 49h), 4Bh

The ranges of n and d vary depending on the selected barcode system.

①

m	Barcode type	Range of k	Range of d (ASCII)
0	UPC-A	$11 \leq k \leq 12$	0 to 9
1	UPC-E	$11 \leq k \leq 12$	0 to 9
2	JAN13 (EAN)	$12 \leq k \leq 13$	0 to 9
3	JAN8 (EAN)	$7 \leq k \leq 8$	0 to 9
4	CODE39	$1 \leq k$	SP,\$,%,+,-,./,0 to 9,A to Z
5	Interleaved 2 of 5 (ITF)	$1 \leq k$ (even number only)	0 to 9
6	NW-7	$1 \leq k$	,\$,+,.,./,:0 to 9,A to D
10	PDF417	$1 \leq k$	1 - 255

②

m	Barcode type	Range of n	Range of d (ASCII)
65	UPC-A	$11 \leq n \leq 12$	0 to 9
66	UPC-E	$11 \leq n \leq 12$	0 to 9
67	JAN13 (EAN)	$12 \leq n \leq 13$	0 to 9
68	JAN8 (EAN)	$7 \leq n \leq 8$	0 to 9
69	CODE39	$1 \leq n \leq 255$	SP,\$,%,+,-,./,0 to 9,A to Z
70	Interleaved 2 of 5 (ITF)	$1 \leq n \leq 255$ (even number only)	0 to 9
71	NW-7	$1 \leq n \leq 255$	,\$,+,.,./,:0 to 9,A to D
72	CODE93	$1 \leq n \leq 255$	0x00 to 0x7F
73	CODE128	$2 \leq n \leq 255$	0x00 to 0x7F
75	PDF417	$1 \leq k$	0 - 255

Explanation

①

- (1) This command ends with the [NUL] code.
- (2) For UPC-A and UPC-E, upon the input of 12-byte barcode data, a barcode is printed. The data subsequent to it is processed as normal data.
- (3) For JAN13, upon the input of 13-byte barcode data, a barcode is printed. The data subsequent to it is processed as normal data.
- (4) For JAN8, upon the input of 8-byte barcode data, a barcode is printed. The data subsequent to it is processed as normal data.
- (5) The number of data items of ITF barcode must be an even number. If it is an odd number, the final data will be ignored.

②

- (1) n refers to the number of data items. n bytes from the subsequent data are processed as barcode data.
- (2) If n is out of the defined range, command processing will be aborted, and the subsequent data will be processed as normal data.

[In standard mode]

- (1) If d is out of the defined range, only paper feed will be performed, and the subsequent data will be processed as normal data.
- (2) If the width of a barcode exceeds the print area per line, it will not be guaranteed that the barcode is printed properly.
- (3) Paper will be fed over the length equivalent to the height of a barcode (including HRI characters if HRI character printing is specified), regardless of a line feed length defined by the 34-dot Line Feed Length Set Command ([ESC] 2) or Minimum Dots for Line Feed Length Set Command ([ESC] 3).
- (4) This command will be valid only if no data is present in the print buffer. If data is present in the print buffer, the data subsequent to m will be processed as normal data.
- (5) After a barcode is printed, the next print position is the beginning of a line.
- (6) There is no impact upon print modes (bold printing, double printing, underlining, and character size) except for inverted printing.

[In page mode]

- (1) A barcode is expanded, but not printed. After the expansion of the barcode, the dot subsequent to the final barcode data is the data expansion starting position.
- (2) If d is out of the defined range, command processing will be aborted, and the subsequent data will be processed as normal data. Note that the data expansion starting position remains unchanged.
- (3) If the width of a barcode exceeds the print area per line, it will not be guaranteed that the barcode is printed properly.

[CODE128]

- (1) Please refer to "APPENDIX: BARCODE" for the overview of CODE 128.
- (2) When this printer is used to print a CODE128 barcode, care must be taken regarding the following points when transmitting barcode data:
 - ① The beginning of the barcode data must always be a code set (CODE A, CODE B, or CODE C), based on which the first code set is selected.
 - ② Special characters are specified by two characters: '{' followed by 1

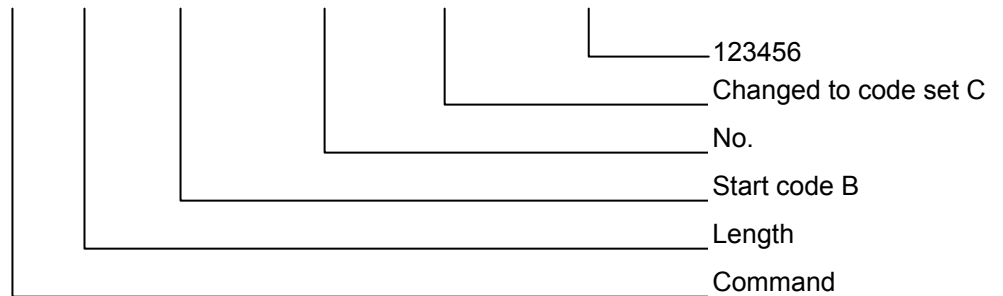
character. Also, an ASCII character '{' is specified by transmitting 2 '{' characters in a row.

Special character	Data to be transmitted	
	ASCII	Hex
SHIFT	{S	7B 53
CODE A	{A	7B 41
CODE B	{B	7B 42
CODE C	{C	7B 43
FNC1	{1	7B 31
FNC2	{2	7B 32
FNC3	{3	7B 33
FNC4	{4	7B 34
'{'	{{	7B 7B

[Example] Data to be transmitted to print “No.123456”

In this example, first, code set B is used to print “No.” After that, the code set is changed to code set C, and the 6-digit number is printed.

[GS] k [49] [0A] [7B] [42] [4E] [6F] [2E] [7B] [43] [0C] [22] [38]



- (3) If the beginning of the barcode data string is not a code set, command processing will be aborted immediately, and the subsequent data will be processed as normal data.
- (4) If the combination of '{' and the following character does not match any special characters, command processing will be aborted immediately, and the subsequent data will be processed as normal data.
- (5) If a character that cannot be used with a selected code set is received, command processing will be aborted immediately, and the subsequent data will be processed as normal data.

- (6) An HRI (Human Readable Interpretation) character corresponding to a shift character and code set character is not printed.
- (7) An HRI character for a function character is space.
- (8) An HRI character for a control character (00H to 1FH and 7FH) is space.

[CODE93]

The B-FP series does not support the following functions:

- (1) An HRI character for a start character (□) is printed at the beginning of the HRI character string.
- (2) An HRI character for a stop character (□) is printed at the end of the HRI character string.
- (3) An HRI character for a control character (00H to 1FH and 7FH) is printed with ■ and an alphabet combined.

Control character		HRI character	Control character		HRI character
ASCII	Hex		ASCII	Hex	
NUL	00	■U	DLE	10	■P
SOH	01	■A	DC1	11	■Q
STX	02	■B	DC2	12	■R
ETX	03	■C	DC3	13	■S
EOT	04	■D	DC4	14	■T
ENQ	05	■E	NAK	15	■U
ACK	06	■F	SYN	16	■V
BEL	07	■G	ETB	17	■W
BS	08	■H	CAN	18	■X
HT	09	■I	EM	19	■Y
LF	0A	■J	SUB	1A	■Z
VT	0B	■K	ESC	1B	■A
FF	0C	■L	FS	1C	■B
CR	0D	■M	GS	1D	■C
SO	0E	■N	RS	1E	■D
SI	0F	■O	US	1F	■E
			DEL	7F	■T

Reference

HRI Character Print Position Set Command ([GS] H)

Barcode Height Set Command ([GS] h)

Barcode Width Set Command ([GS] w)

6.47 APPENDIX : BARCODE

6.47.1 CODE128

Code set A: ASCII characters ranging between 00H and 5FH can be represented.

Code set B: ASCII characters ranging between 20H and 7FH can be represented.

Code set C: Two digits can be represented using 1 character.

In addition to the above characters, CODE128 has the following special characters:

- Shift character (SHIFT)

In code set A, a character immediately after SHIFT is treated as a character of code set B. In code set B, a character immediately after SHIFT is treated as a character of code set A. Note that SHIFT cannot be used in code set C.

- Code set character (CODEA, CODEB, CODEC)

The code set subsequent to a code set character is changed to A, B, or C.

- Function character (FNC1, FNC2, FNC3, FNC4)

Uses of the function character vary depending on the application. Note that only FNC1 can be used in code set C.

[Characters printable with code set A]

Char	Data to be transmitted
NUL	00
SOH	01
STX	02
ETX	03
EOT	04
ENQ	05
ACK	06
BEL	07
BS	08
HT	09
LF	0A
VT	0B
FF	0C
CR	0D
SO	0E
SI	0F
DLE	10
DC1	11
DC2	12
DC3	13
DC4	14
NAK	15
SYN	16
ETB	17

Char	Data to be transmitted
CAN	18
EM	19
SUB	1A
ESC	1B
FS	1C
GS	1D
RS	1E
US	1F
SP	20
!	21
"	22
#	23
\$	24
%	25
&	26
'	27
(	28
)	29
*	2A
+	2B
,	2C
-	2D
.	2E
/	2F

Char	Data to be transmitted
0	30
1	31
2	32
3	33
4	34
5	35
6	36
7	37
8	38
9	39
:	3A
;	3B
<	3C
=	3D
>	3E
?	3F
@	40
A	41
B	42
C	43
D	44
E	45
F	46
G	47

Char	Data to be transmitted
H	48
I	49
J	4A
K	4B
L	4C
M	4D
N	4E
O	4F
P	50
Q	51
R	52
S	53
T	54
U	55
V	56
W	57
X	58
Y	59
Z	5A
[	5B
\	5C
]	5D
~	5E
	5F

Char	Data to be transmitted
FNC1	7B 31
FNC2	7B 32
FNC3	7B 33
FNC4	7B 34
SHIFT	7B 35
CODEB	7B 42
CODEC	7B 43

[Characters printable with code set B]

Char	Data to be transmitted
SP	20
!	21
"	22
#	23
\$	24
%	25
&	26
'	27
(	28
)	29
*	2A
+	2B
,	2C
-	2D
.	2E
/	2F
0	30
1	31
2	32
3	33
4	34
5	35
6	36
7	37

Char	Data to be transmitted
8	38
9	39
:	3A
;	3B
<	3C
=	3D
>	3E
?	3F
@	40
A	41
B	42
C	43
D	44
E	45
F	46
G	47
H	48
I	49
J	4A
K	4B
L	4C
M	4D
N	4E
O	4F

Char	Data to be transmitted
P	50
Q	51
R	52
S	53
T	54
U	55
V	56
W	57
X	58
Y	59
Z	5A
[	5B
\	5C
]	5D
~	5E
`	5F
a	61
b	62
c	63
d	64
e	65
f	66
g	67

Char	Data to be transmitted
h	68
i	69
j	6A
k	6B
l	6C
m	6D
n	6E
o	6F
p	70
q	71
r	72
s	73
t	74
u	75
v	76
w	77
x	78
y	79
z	7A
{	7B
	7C
}	7D
~	7E
DEL	7F

Char	Data to be transmitted
FNC1	7B 31
FNC2	7B 32
FNC3	7B 33
FNC4	7B 34
SHIFT	7B 53
CODEA	7B 41
CODEC	7B 43

[Characters printable with code set C]

Char	Data to be transmitted
00	00
01	01
02	02
03	03
04	04
05	05
06	06
07	07
08	08
09	09
10	0A
11	0B
12	0C
13	0D
14	0E
15	0F
16	10
17	11
18	12
19	13
20	14
21	15
22	16
23	17

Char	Data to be transmitted
24	18
25	19
26	1A
27	1B
28	1C
29	1D
30	1E
31	1F
32	20
33	21
34	22
35	23
36	24
37	25
38	26
39	27
40	28
41	29
42	2A
43	2B
44	2C
45	2D
46	2E
47	2F

Char	Data to be transmitted
48	30
49	31
50	32
51	33
52	34
53	35
54	36
55	37
56	38
57	39
58	3A
59	3B
60	3C
61	3D
62	3E
63	3F
64	40
65	41
66	42
67	43
68	44
69	45
70	46
71	47

Char	Data to be transmitted
72	48
73	49
74	4A
75	4B
76	4C
77	4D
78	4E
79	4F
80	50
81	51
82	52
83	53
84	54
85	55
86	56
87	57
88	58
89	59
90	5A
91	5B
92	5C
93	5D
94	5E
95	5F

Char	Data to be transmitted
96	60
97	61
98	62
99	63
FNC1	7B 31
CODEA	7B 41
CODEB	7B 42

6.47.2 UPC-E

Conversion to UPC-E is performed as shown by the tables below.

UPC-E (7 digit) version								UPC-A (12 digit) version											
P1	P2	P3	P4	P5	P6	P7	P8	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
0	X1	X2	Y1	Y2	Y3	0	CD	0	X1	X2	0	0	0	0	0	Y1	Y2	Y3	CD
0	X1	X2	Y1	Y2	Y3	1	CD	0	X1	X2	1	0	0	0	0	Y1	Y2	Y3	CD
0	X1	X2	Y1	Y2	Y3	2	CD	0	X1	X2	2	0	0	0	0	Y1	Y2	Y3	CD
0	X1	X2	X3	Y1	Y2	3	CD	0	X1	X2	X3	0	0	0	0	0	Y1	Y2	CD
0	X1	X2	X3	X4	Y1	4	CD	0	X1	X2	X3	X4	0	0	0	0	0	Y1	CD
0	X1	X2	X3	X4	X5	Y1	CD	0	X1	X2	X3	X4	X5	0	0	0	0	Y1	CD

- Data check is performed for the “0” portion (shaded in the right table) to categorize the data into 4 patterns from ① to ④. If the data does not match the 4 patterns, command processing will be aborted.
- The data categorized into the 4 patterns from ① to ④ is converted, as shown by the left table.
- A4 data in pattern ① accepts only 0, 1, or 2. However, the printer does not perform any data integrity check. Therefore, a barcode will be printed even if any value other than 0 to 2 is specified for A4 (P7) data.

6.48 Barcode HORIZONTAL SIZE command [GS] w

Function Sets the horizontal size of the barcode.

Format [GS] w n

Term ASCII: GS w
Hex: 1Dh 77h

[Parameter]

n: Horizontal size of the barcode
 $1 \leq n \leq 6$ (01h $\leq$ n $\leq$ 06h)

Default value n : 3

Explanation The maximum value varies depending on model and barcode type.

	Model	B-FP3D-GH/GS
	Range	01H to 06H
	Initial value	03H
UPC-A, UPC-E, JAN13, JAN8, CODE128	Maximum value	06H
CODE39 Interleaved 2 of 5, NW-7	Maximum value	06H

[UPC-A, UPC-E, JAN13, JAN8, or CODE128]

n	1 module		2 modules		3 modules		4 modules	
	Bar	Space	Bar	Space	Bar	Space	Bar	Space
02H	2		4		6		8	
03H	3		6		9		12	
04H	4		8		12		16	
05H	5		10		15		20	
06H	6		12		18		24	
07H	7		14		21		28	
08H	8		16		24		32	
09H	9		18		27		36	

(Unit: dot)

[CODE39, Interleaved 2 of 5, or NW-7]

n	Narrow		Wide		Character-to-character space
	Bar	Space	Bar	Space	
02H	2	2	4	4	2
03H	3	3	6	6	3
04H	4	4	8	8	4
05H	5	5	10	10	5
06H	6	6	12	12	6

(Unit: dot)

* The character-to-character space does not exist in Interleaved 2 of 5.

(1) A multilevel barcode refers to the following barcode system:

UPC-A, UPC-E, JAN13, JAN8, CODE93, and CODE128

(2) A 2-level barcode refers to the following barcode system:

CODE39, ITF, and NW-7

(3) If "n" is out of range, this command will be ignored.

Reference

Barcode Print Command ([GS] k)

6.49 Barcode Height command [GS] h

Function Sets the barcode height.

Format [GS] hn

Term ASCII: GS h
Hex: 1Dh 68h

[Parameter]

n: (n * 1.38) of dots for the barcode height
 $1 \leq n \leq 255$ (01h $\leq$ n $\leq$ FFh)

Default value n: 162

Explanation (1) Below table shows the barcode height in dots based on 'n' value..

n	height in dots	n	height in dots	n	height in dots	n	height in dots
0	1	40	55	80	AB	C0	100
1	1	41	57	81	AC	C1	101
2	3	42	58	82	AD	C2	103
3	4	43	59	83	AF	C3	104
4	5	44	5B	84	B0	C4	105
5	7	45	5C	85	B1	C5	107
6	8	46	5D	86	B3	C6	108
7	9	47	5F	87	B4	C7	109
8	B	48	60	88	B5	C8	10B
9	C	49	61	89	B7	C9	10C
A	D	4A	63	8A	B8	CA	10D
B	F	4B	64	8B	B9	CB	10F
C	10	4C	65	8C	BB	CC	110
D	11	4D	67	8D	BC	CD	111
E	13	4E	68	8E	BD	CE	113
F	14	4F	69	8F	BF	CF	114
10	15	50	6B	90	C0	D0	115
11	17	51	6C	91	C1	D1	117
12	18	52	6D	92	C3	D2	118
13	19	53	6F	93	C4	D3	119
14	1B	54	70	94	C5	D4	11B
15	1C	55	71	95	C7	D5	11C
16	1D	56	73	96	C8	D6	11D
17	1F	57	74	97	C9	D7	11F
18	20	58	75	98	CB	D8	120
19	21	59	77	99	CC	D9	121
1A	23	5A	78	9A	CD	DA	123
1B	24	5B	79	9B	CF	DB	124
1C	25	5C	7B	9C	D0	DC	125
1D	27	5D	7C	9D	D1	DD	127
1E	28	5E	7D	9E	D3	DE	128

1F	29	5F	7F	9F	D4	DF	129
20	2B	60	80	A0	D5	E0	12B
21	2C	61	81	A1	D7	E1	12C
22	2D	62	83	A2	D8	E2	12D
23	2F	63	84	A3	D9	E3	12F
24	30	64	85	A4	DB	E4	130
25	31	65	87	A5	DC	E5	131
26	33	66	88	A6	DD	E6	133
27	34	67	89	A7	DF	E7	134
28	35	68	8B	A8	E0	E8	135
29	37	69	8C	A9	E1	E9	137
2A	38	6A	8D	AA	E3	EA	138
2B	39	6B	08F	AB	E4	EB	139
2C	3B	6C	90	AC	E5	EC	13B
2D	3C	6D	91	AD	E7	ED	13C
2E	3D	6E	93	AE	E8	EE	13D
2F	3F	6F	94	AF	E9	EF	13F
30	40	70	95	B0	EB	F0	140
31	41	71	97	B1	EC	F1	141
32	43	72	98	B2	ED	F2	143
33	44	73	99	B3	EF	F3	144
34	45	74	9B	B4	F0	F4	145
35	47	75	9C	B5	F1	F5	147
36	48	76	9D	B6	F3	F6	148
37	49	77	9F	B7	F4	F7	149
38	4B	78	A0	B8	F5	F8	14B
39	4C	79	A1	B9	F7	F9	14C
3A	4D	7A	A3	BA	F8	FA	14D
3B	4F	7B	A4	BB	F9	FB	14F
3C	50	7C	A5	BC	FB	FC	150
3D	51	7D	A7	BD	FC	FD	151
3E	53	7E	A8	BE	FD	FE	153
3F	54	7F	A9	BF	FF	FF	154

Reference

Barcode Print Command ([GS] k)

6.50 NUMERALS UNDER BARS COMMAND [GS] H

Function	Specifies a print position of an HRI character when a barcode is printed.		
Format	[GS] Hn		
Term	ASCII: GS H Hex: 1Dh 48h		
	[Parameter]		
	n: Print position of HRI character	$0 \leq n \leq 3$ (00h $\leq n \leq$ 03h) or $48 \leq n \leq 50$ (30h $\leq n \leq$ 33h)	
	0: Not printing		
	1: Above barcode		
	2: Below barcode		
	3: Above and below barcode		
Default value	n: 0		
Explanation	(1) HRI stands for Human Readable Interpretation.		
Reference	Barcode Print Command ([GS] k)		

6.51 QRcode : select the model [GS] (k

Function	Selects the QR code model
----------	---------------------------

Format	[GS] (k pL pH cn fn n1 n2
--------	----------------------------

Term	ASCII: GS (k Hex: 1Dh 28h 6Bh
------	-----------------------------------

[Parameter]

pL, pH : number of bytes after pH

cn : 49

fn : 65

n1 : 49 = model 1 code conversion processing

50 = model 2 conversion processing

200 = Micro QR Code

n2 : 0

Default value	n1 = 50, n2 = 0
---------------	-----------------

6.52 Qrcode : set the size of module [GS] (k

Function	Sets the size of the module for QR Code as n dots
----------	---

Format	[GS] (k pL pH cn fn n
--------	------------------------

Term	ASCII: GS (k Hex: 1Dh 28h 6Bh
------	-----------------------------------

[Parameter]

pL, pH : number of bytes after pH

cn : 49

fn : 67

n : 1 - 16

Default value	n : 4
---------------	-------

Explanation

- (1) User must secure the quiet zone (left, right, upward, and downward space areas defined by the QR Code symbol specification) for QR code printing.
- (2) The command is ignored if n is out of range.

6.53 QRcode – select the error correction level

[GS] (k

Function	Selects the error correction level for QR code		
Format	[GS] (k pL pH cn fn n		
Term	ASCII: GS (k Hex: 1Dh 28h 6Bh [Parameter] pL, pH : number of bytes after pH cn : 49 fn : 69 n : 48 = Select error correction level L 7% n : 49 = Select error correction level M 15% n : 50 = Select error correction level Q 25% n : 51 = Select error correction level H 30% When model1 or model2 selected n=48,49,50,51 When Micro QR selected n=48,49,50		
Default value	n : 48		

6.54 QRcode – store the data in symbol storage area [GS] (k

Function	Stores the QR Code symbol data(d1...dk) into the symbol storage area(RAM)
Format	[GS] (k pL pH cn fn n d1...dk
Term	ASCII: GS (k Hex: 1Dh 28h 6Bh [Parameter] pL, pH : number of bytes after pH cn : 49 fn : 80 n : 48 d : 0 – 255 (data to store in symbol storage area) k : (pL + pH * 256) – 3
Explanation	(1) Stores the QR Code symbol data into the storage area (RAM)
Reference	QR Code : prints the data in symbol storage area (1Dh 28h 6Bh)

6.55 QRcode – prints the data in symbol storage area [GS] (k

Function	Encodes and prints the QR Code symbol data in the symbol storage area		
Format	[GS] (k pL pH cn fn n		
Term	ASCII: GS (k Hex: 1Dh 28h 6Bh [Parameter] pL, pH : number of bytes after pH cn : 49 fn : 81 n : 48		
Explanation	(1)	Encodes and prints the QR code symbol data in the symbol storage area	

6.56 QRcode – Transmits the size information of the SYMBOL data in symbol storage area [GS] (k

Function	Printer transmits the size information for the encoded QR Code symbol data in the symbol storage area
Format	[GS] (k pL pH cn fn n
Term	ASCII: GS (k Hex: 1Dh 28h 6Bh [Parameter] pL, pH : number of bytes after pH cn : 49 fn : 82 n : 48
Explanation	

- (1) Transmits the size information for encoded QR Code symbol data into the storage area as below mentioned.

Description	Hex	Decimal	Data
Header	37	55	1 byte
Identifier	36	54	1 byte
Horizontal size	30-39	48-57	1 - 5 bytes
Separator	1F	31	1 byte
Vertical Size	30-39	48-57	1 - 5 bytes
Separator	1F	31	1 byte
Fixed Value	31	49	1 byte
Separator	1F	31	1 byte
Other information	30 or 31	48 or 49	1 byte
NULL	00	0	1 byte

Note:

In the above, "Other information" represents the possibility of printing QR Code.

Other Information: 0x30 - Printing is possible

0x31 - Printing is impossible

Horizontal and vertical sizes are specified as ASCII value of received byte. They can be obtained by following equations.

Horizontal size = Number of cells in Horizontal Direction × Symbol size specified by the command 1D 28 6B 03 00 31 43 n

Vertical size = Number of cells in Vertical Direction × Symbol size specified by the command 1D 28 6B 03 00 31 43 n

Example: If Symbol size is specified as 10 by the command of 1D 28 6B 03 00 31 43 n and number of pixel in horizontal direction is 21 then horizontal size will be 10 * 21=210. So the printer's output will be 37 36 32 31 30 1f 32 31 30 1f 31 1f 30 00.

6.57 Print Area Width Set command [GS] W

Function Specifies the width of the print area defined by nL and nH.

The width of the print area is determined by $[(nL + nH \times 256) \text{ dots}]$.

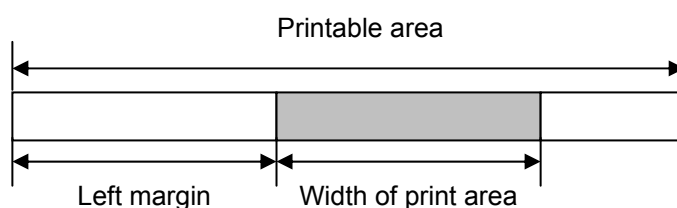
Format [GS] W nL nH

Term ASCII: GS W
Hex: 1Dh 57h

[Parameter]

nL: Width of print area $0 \leq nL \leq 255$ (00h $\leq$ nL $\leq$ FFh)

nH: Width of print area $0 \leq nH \leq 255$ (00h $\leq$ nH $\leq$ FFh)



Default value 576 dots (nL = 40, nH = 02) (B-FP3D-GH/GS)

Explanation

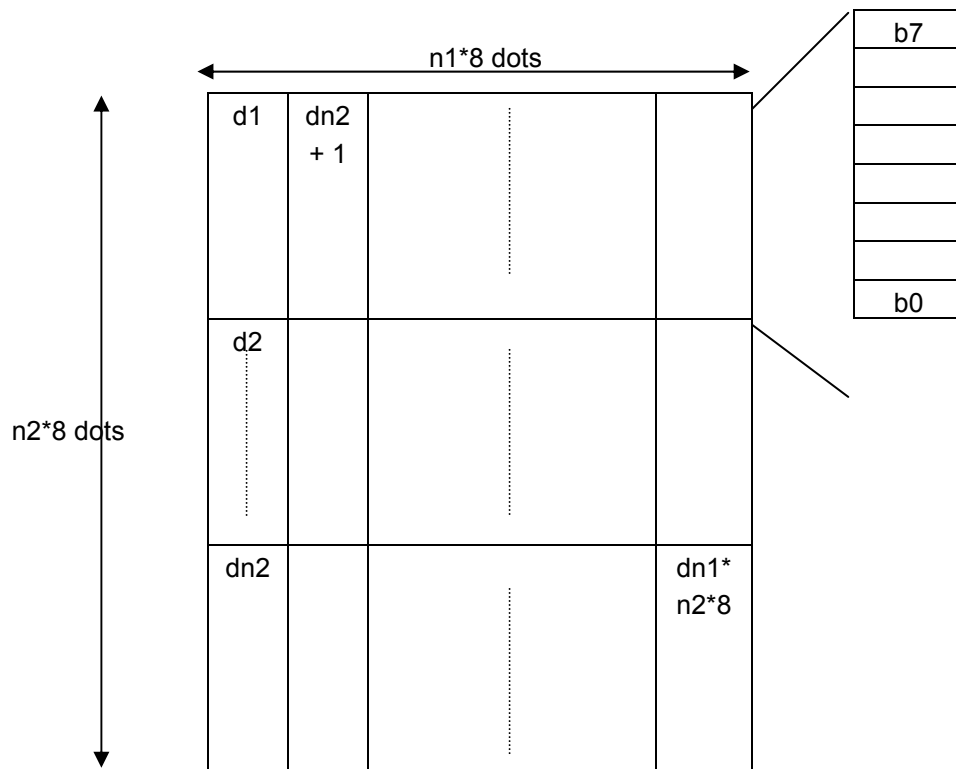
- (1) This command will be executed only if received at the beginning of a line.
- (2) If this command is received while the printer is in page mode, the defined parameter setting will be applied after switching to standard mode.
- (3) The command settings do not have any impacts upon page mode.
- (4) If a value beyond the printable area per line is inputted, all area except for the left margin will be specified as a print area.
- (5) If a value specified for the width of the print area is smaller than the width of a character to be printed, printing will only be performed by up to the specified width of the print area.
- (6) If the width of the print area is set to 0 ([GS] W [00] [00]), printing will start from the left edge of the printable area.

Reference

Left Margin Set Command ([GS] L)

6.58 Downloaded Bit image Define command [GS] *

Function	Defines a downloaded bit image that contains the number of dots specified by n1 and n2.
Format	[GS] * n1 n2 [d ₁ d ₂ d ₃ ...d _(n1×n2×8)]
Term	ASCII: GS * Hex: 1Dh 2Ah [Parameter] n1: No. of dots in the horizontal direction (n1 × 8) Font Size 1 1 ≤ n1 ≤ 72 (01h ≤ n1 ≤ 48h) Font Size 2 1 ≤ n1 ≤ 64 (01h ≤ n1 ≤ 40h) n2: No. of dots in the vertical direction (n2 × 8) 1 ≤ n2 ≤ 64 (01h ≤ n2 ≤ 40h) * n1 × n2 ≤ 1,536 d: Image data 1 ≤ d ≤ 255 (01h ≤ d ≤ FFh) * 1 refers to a bit to be printed, and 0 to the one not to be printed.
Default value	
Explanation	(1) If n1 × n2 is out of the defined range, this command will be ignored. (2) d refers to bit image data. A bit corresponding to a dot to be printed is set to 1, and the one corresponding to a dot not to be printed is set to 0. (3) The defined downloaded bit image is cleared in the event of: Initialize Command ([ESC] @) Downloaded Character (Writable Character) Define Command ([ESC] &) Resetting or turning the printer off (4) A downloaded bit image is defined. (5) A specified logo is valid until resetting or the printer is turned off. However, in the event of a parameter error (e.g. a value out of range), the most recently stored logo is not guaranteed. (6) At a moment when an error occurs with parameter n1 or n2 due to a factor such as an unmatched condition, command processing will be aborted, and the data subsequent to n2 will be processed as normal data. (7) This command is valid in page mode only when the parameter of the Page Mode Print Direction Set Command ([ESC] T) is set to 0 (0-degree rotation). The bit image will not be rotated if any value other than 0 is specified. (8) The figure on the next page illustrates a relationship between data and printed dots.



Reference

Downloaded Bit image Print Mode Set Command ([GS] /)

6.59 Downloaded Bit image Print Mode Set command [GS] /

Function Prints a logo (image data) specified by n in mode specified by k

Format [GS] /m

Term ASCII: GS /
Hex: 1Dh 2Fh

[Parameter]

m: Downloaded bit image data print mode $0 \leq m \leq 3$ (00h $\leq m \leq$ 03h)

or

$48 \leq m \leq 51$ (30h $\leq m \leq$ 33h)

0,48	Normal mode
1,49	Double-width mode
2,50	Double-height mode
3,51	Quadruple mode

- Explanation**
- (1) Unless downloaded bit image data is defined, this command will be ignored.
 - (2) When standard mode is selected, the command will be valid only if no data is present in the print buffer.
 - (3) Inverted printing, bold printing, double printing, underlining, character size, or reversed printing will become invalid even if specified.
 - (4) If there is a line where the print area specified by the Left Margin Set Command ([GS] L) or Print Area Width Set Command ([GS] W) does not reach the minimum width in bit image mode (1 dot in normal or double-height mode, or 2 dots in double-width or quadruple mode), the print area will be extended to the right until it reaches the minimum width in bit image mode to the extent that the printable area is not exceeded. This only applies to such a line.

Reference Downloaded Bit image Define Command ([GS] *)

6.60 Macro Definition Start/End command [GS] :

Function	Specifies the start or end of a macro.
Format	[GS] :
Term	ASCII: GS : Hex: 1Dh 3Ah
Explanation	(1) If this command is included under normal operation, the start of a macro definition will be specified. If this command is included while a macro is being defined, the end of the macro definition will be specified. (2) If the Macro Execute Command ([GS] ^) is included while the macro is being defined, the macro definition will be aborted, and cleared. (3) A macro is "undefined" initially. (4) Undefined items are not cleared by the Initialize Command ([ESC] @). (5) If [GS] : is included immediately after the receipt of this command, a macro will become "undefined." (6) The maximum permissible number of bytes for a macro definition will be limited to 2,048, even if it exceeds 2,048.
Reference	Macro Execute Command ([GS] ^)

6.61 Macro Execute command [GS] ^

Function	Executes a defined macro.
Format	[GS] ^ n1 n2 n3
Term	ASCII: GS ^ Hex: 1Dh 5Eh [Parameter] n1: No. of times macro is executed $0 \leq n1 \leq 255$ (00h $\leq$ n1 $\leq$ FFh) n2: Ignored. n3: Mode to execute a macro $0 \leq n3 \leq 1$ (00h $\leq$ n3 $\leq$ 01h) 0: A macro is executed n1 times. 1: After the first receipt print, the STATUS LED (green) blinks and waits for the [FEED] key to be pressed. Once the [FEED] key is pressed, a macro is executed once. This is repeated n1 times.
Explanation	(1) If this command is included while the macro is being defined, the macro definition will be aborted, and cleared. (2) Nothing will be done with no macro defined or with n1 = 0. (3) The [FEED] key is inoperable during the specified waiting time when a macro is executed.
Reference	Macro Definition Start/End Command ([GS] :)

6.62 Left Margin Set command [GS] L

Function Sets the left margin as specified by nL and nH.

The left margin is determined by $[(nL + nH \times 256) \times \text{dots}]$.

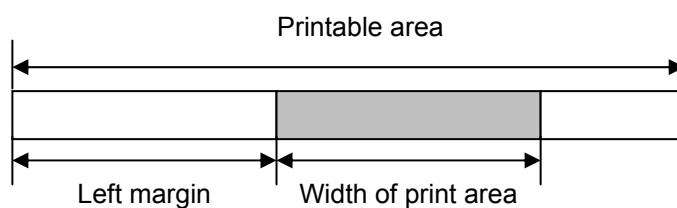
Format [GS] LnLnH

Term ASCII: GS L
Hex: 1Dh 4Ch

[Parameter]

nL: Width of print area $0 \leq nL \leq 255$ (00h $\leq$ nL $\leq$ FFh)

nH: Width of print area $0 \leq nH \leq 255$ (00h $\leq$ nH $\leq$ FFh)



Default value nL = 0, nH = 0

- Explanation**
- (1) This command will be valid only if received at the beginning of a line.
 - (2) If this command is received while the printer is in page mode, the defined parameter setting will be applied after switching to standard mode.
 - (3) The command settings do not have any impacts upon page mode.
 - (4) The maximum permissible left margin is equal to the width of the printable area. If a specified value exceeds the maximum value, it will be rounded to the maximum value.

Reference Print Area Width Set Command ([GS] W)

6.63 Character Size Set command [GS] !

Function Specifies a character size (magnification in the vertical / horizontal direction).

Format [GS] ! n

Term ASCII: GS !
Hex: 1Dh 21h

[Parameter]

n: Specifying a character size $1 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh)

* $1 \leq$ Magnification in vertical direction ≤ 8 , $1 \leq$ Magnification in horizontal direction ≤ 8

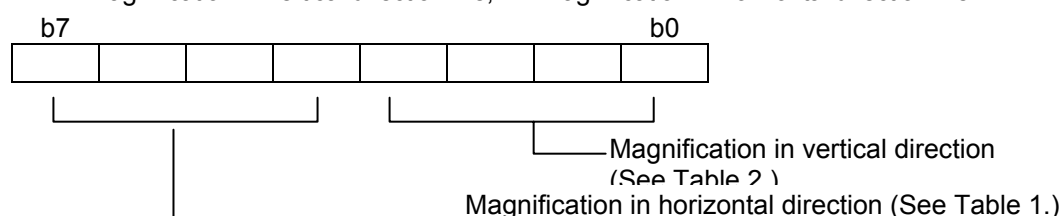


Table 1 [Magnification in horizontal direction]

1-byte data (hex)	Magnification
00	x 1 (standard)
10	x 2 (double width)
20	x 3
30	x 4
40	x 5
50	x 6
60	x 7
70	x 8

Table 2 [Magnification in vertical direction]

1-byte data (hex)	Magnification
00	x 1 (standard)
01	x 2 (double height)
02	x 3
03	x 4
04	x 5
05	x 6
06	x 7
07	x 8

Default value n: 0

Example A magnification in the horizontal direction is x 3, and a magnification in the vertical direction is x 6.

The command is [GS] ! [25].

- Explanation**
- (1) This parameter setting applies to all characters (ANK and Chinese characters) except for HRI characters.
 - (2) If either a specified magnification in the vertical or horizontal direction is out of the defined range, this command will be ignored.
 - (3) In standard mode, "vertical direction" refers to the paper feed direction, and "horizontal direction" to the direction perpendicular to the paper feed direction. Therefore, if 90-degree rotation is specified, this relationship will be reversed.
 - (4) In page mode, "vertical direction" refers to the vertical direction of a character, and the horizontal direction to the width.
 - (5) Double width and double height can be set or canceled by the Print Mode All Set Command ([ESC] !). But the most recently processed command is applied.

Reference Print Mode All Set Command ([ESC] !)

6.64 SET HORIZONTAL AND VERTICAL MINIMUM MOTION UNIT [GS] P

Function	Sets the horizontal and vertical motion units to 1/x inch and 1/y inch respectively
Format	[GS] P x y
Term	ASCII: GS P Hex: 1Dh 50h [Parameter] x: Horizontal minimum motion unit $0 \leq nL \leq 255$ (01h $\leq$ nL $\leq$ FFh) y: Vertical minimum motion unit $0 \leq nH \leq 255$ (01h $\leq$ nH $\leq$ FFh)
Default value	x: 203 Y: 203
Explanation	(1) When x or y is set to 0, the default setting for that motion unit is used.

6.65 SET COLUMN [ESC] DC4

Function	Sets the Column of the first character
Format	[ESC] DC4 n
Term	ASCII: ESC DC4 Hex: 1Bh 14h [Parameter] FONT A: $1 \leq n \leq 48$ (01h $\leq n \leq$ 30h) FONT B: $1 \leq n \leq 64$ (01h $\leq n \leq$ 40h) FONT A: $1 \leq n \leq 42$ (01h $\leq n \leq$ 2Ah) FONT B: $1 \leq n \leq 56$ (01h $\leq n \leq$ 38h)
Default value	n: 1
Explanation	(1) Prints the first character of the next print line in column n. (2) It must be sent for each line not printed at column one.

6.66 SELECT CHARACTER FONT [ESC] M

Function	Selects the character font type.
Format	[ESC] M n
Term	ASCII: ESC M Hex: 1Bh 4Dh [Parameter] 0, 48 (00h, 30h) = FONT A 1, 49 (01h, 31h) = FONT B 59 (3Bh) = MODE SELECT COMMAND. See (2) below for further information
Default value	n: 0
Explanation	(1) This command selects character font same as bit 0 of select print mode command. (2) Please refer TEC ORIGINAL COMMAND section for the mode change command. (3) If parameter is other than accepted values, Ignores the command.
Reference	Select Print Mode Command (1Bh 21h)

6.67 PRINT RASTER GRAPHICS

[ESC] ,

Function	Prints the raster graphics.
Format	[ESC] , c mL, mH, d1d2 d3.....d80
Term	ASCII: ESC , Hex: 1Bh 2Ch

[Parameter]
C : 00H Black only
01H Red only
02H Black and Red only
mL, mH is vertical dot count
d1d2 d3...d80 data bytes.

Explanation	(1) This command prints the raster graphics. (2) 80 bytes of the data should send for printing one line raster graphics.
-------------	---

	d1	d2			d79	d80	
Line1							
Line2							

6.68 DOWNLOAD BMP LOGO [US] (*.BMP file)

Function	Stores the BMP file into RAM or FLASH.
Format	[US] (+ *.BMP)
Term	ASCII: US(+ *.BMP) Hex: 1Fh(+ *.BMP)
Explanation	(1) This command is used by sending the file data of a monochrome BMP file preceded by a 0x1F. (2) Maximum width of the image is 576 dots. (3) Maximum height of the image is 512 dots. (4) The bitmap is stored in the printer in the same manner as s downloaded bit image. (5) The downloaded BMP file can be printed by using the Print Downloaded bit image(1D 2F m) (6) BMP file images that are not monochrome are ignored.
Reference	Select Memory Type to Save Logos (1F 33 n) Print Downloaded Bit image (1D 2F m)

6.69

SELECT CURRENT LOGO

[US] 0

Function	Selects a logo to be defined or printed. The active logo n remains in use until this command is sent again with a different logo n
Format	[US] 0 n
Term	ASCII: US 0 Hex: 1Fh 30h [Parameter] n: $0 \leq n \leq 255$ ($00h \leq n \leq FFh$)
Default value	$n = 0$
Explanation	(1) When this command precedes a logo definition, that definition is stored in Flash Memory as logo n. (2) If there is already a different definition in Flash Memory for logo n, the first is inactivated and the new definition is used. The inactive definition is not erased from Flash and continues to take up space in Flash Memory (3) When this command precedes a logo print command and n is different from the previously active logo selected, the printer retrieves the logo definition for n from Flash Memory and prints it. If there is no definition for logo n, then no logo is printed. (4) The application using multiple logos, writing a user-defined character set into Flash Memory, or both, is responsible for erasing the logo and user-defined character set Flash Memory page when the logo area is full or before a new character set is defined.
Reference	Select Memory Type to Save Logos (1F 33 n) Print Downloaded Bit image (1D 2F m) Download BMP logo (1F +(*.BMP))

6.70 USER NV MEMORY BIT IMAGE PRINTING [FS] P

Function	Print NV bit image 'n' in "m" mode		
Format	[FS] p n m		
Term	ASCII: FS p Hex: 1Ch 70h		
	[Parameter]		
	n: NV bit image number	$0 \leq n \leq 255$ (00h $\leq n \leq$ FFh)	
	m: bit image mode	$0 \leq m \leq 3$ (00h $\leq m \leq$ 03h); $48 \leq m \leq 51$ (30h $\leq m \leq$ 33h);	
Explanation	(1) NV bit image is specified by FS q on a flash memory and printed by FS p (2) When undefined NV bit image 'n' is selected, this command is ignored (3) When the printer is in the standard mode and no data is in the print buffer, this command is valid (4) Emphasis, double strike, underline, font size, white/black reverse printing, rotate mode do not affect to the NV bit image printing. (5) When the specified size of NV bit image is larger than the printable area, outside image of printable area is not printed. (6) Line pitch commands (ESC 2, ESC 3) do not affect to the NV bit image printing. (7) After printing NV bit image, next print position is the beginning of the line.		
Reference	User NV memory bit image define (1Fh 71h)		

6.71 USER NV MEMORY BIT IMAGE DEFINE [FS] q

Function	Defines NV bit image 'n'.
Format	[FS] q n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n
Term	ASCII: FS q Hex: 1Ch 71h [Parameter] n: NV bit image number $0 \leq n \leq 255$ (00h $\leq$ n $\leq$ FFh) xL, xH : width of NV image $0 \leq (xL + xH \times 256) \leq 576$ (00h $\leq$ (xL + xH $\times$ 256) $\leq$ 240h) yL, yH : height of NV image $0 \leq (xL + xH \times 256) \leq 288$ (00h $\leq$ (xL + xH $\times$ 256) $\leq$ 120h)
Explanation	(1) This command is valid when it is processed at the beginning of a line in the standard mode. (2) This command is valid after processing the first 7 bytes (Fs 1 n xL xH yL yH). (3) When the NV bit image size specified xL, xH, yL and yH is larger than the empty area, this command is processed as the parameter error. This command is ignored when it is the parameter error. (4) When the parameter error is detected after definition of second image, the definition of the following image is ignored but the definitions of the previous image are valid. (5) "d" means bit image. black dot is "1" and white dot is "0". (6) One definition of NV bit image consist from xL xH yL yH d1....dk. (7) The definitions of NV bit image queue based on the bit image number. (8) During writing data to the flash memory, the printer issues the busy signal (9) If the printer receives this command during the MACRO definition, the printer cancels the MACRO definition. (10) The specified NV bit image are not initialized by ESC@ command, reset and power-off
Reference	User NV memory bit image Print (1Fh 70h)

6.72 SET Graphics data [GS] (L

Function Sets the Graphics data

Format [GS] (L pL pH m fn [parameter]

Term ASCII: GS (L
Hex: 1Dh 28h 4Ch

[Parameter]

pL pH : number of bytes after pH. $0 \leq (pL + pH * 256) \leq 65535$
(00h $\leq$ (pL + pH * 256) $\leq$ FFFFh)

Function Sets the Graphics data

Format [GS] 8 L pL pH m fn [parameter]

Term ASCII: GS 8 L
Hex: 1Dh 38h 4Ch

[Parameter]

p1 p2 p3 p4 : number of bytes after p4.
 $0 \leq (p1 + p2 * 256 + p3 * 65536 + p4 * 16777216) \leq 65535$
(00h $\leq$ (p1 + p2 * 256 + p3 * 65536 + p4 * 16777216) $\leq$ FFFFFFFFh)

Explanation

The only difference between GS (L and GS 8 L is as listed below:

Command	Parameter	Structure	Maximum value	Description
GS (L	pL, pH	2 bytes	65,535	Parameter value is 2 bytes less than that for GS 8 L. Used to fix the parameter value. Used when sending data divided into blocks.
GS 8 L	p1,p2,p3,p4	4 bytes	4,294,967,295	Possesses powerful range of expression. Used for batch transfer of large volumes of data.

Function code (fn) specifies the function.

fn	Format	Function No.	Function name
0, 48	GS (L pL pH m fn	48	Transmit the NV graphics memory capacity
2, 50	GS (L pL pH m fn	50	Print the graphics data in the print buffer.
3, 51	GS (L pL pH m fn	51	Transmit the remaining capacity of the NV graphics memory.
64	GS (L pL pH m fn d1 d2	64	Transmit the key code list for defined NV graphics.
65	GS (L pL pH m fn d1 d2 d3	65	Delete all NV graphics data.
66	GS (L pL pH m fn kc1 kc2	66	Delete the specified NV graphics data.
67	GS (L pL pH m fn a kc1 kc2 b	67	Define the NV graphics data (raster

	xL xH yL yH [c d1...dk]1... [c d1...dk]b GS 8 L p1 p2 p3 p4 m fn a kc1 kc2 b xL xH yL yH [c d1...dk]1... [cd1...dk]b		format).
69	GS (L pL pH m fn kc1 kc2 x y	69	Print the specified NV graphics data.
112	GS (L pL pH m fn a bx by c xL xH yL yH d1...dk GS 8 L p1 p2 p3 p4 m fn a bx by c xL xH yL yH d1...dk	112	Store the graphics data in the print buffer (raster format).

GS (L command set is the new set of commands to replace the existing logo commands(1C 70..., 1D 2A..., 1F "BMP"). Once GS (L is used the existing logos defined using the old logo commands will be erased. This is applicable to NVROM and RAM. If existing logo commands (1D 2A..., 1F "BMP"...) are received after GS (L then these commands will be ignored.

Switch

If user wants to use the existing logo commands, then 1C 70 should be used first to define the logo commands. 1C 70 command erases the logos defined by GS (L commands. Then the other logo commands (1D 2A..., 1F "BMP"...) can be used.

6.73

Transmit the NV graphics memory capacity (Function 48)

Function	Transmits the entire capacity of the NV graphics area		
Format	[GS] (L pL pH m fn		
Term	ASCII: GS (L Hex: 1Dh 28h 4Ch [Parameter] pL pH : number of bytes after pH. $0 \leq (pL + pH * 256) \leq 2$ $(00h \leq (pL + pH * 256) \leq 02h)$		
Default value	pL = 2 pH = 0 m = 48 fn = 0, 48		

Explanation

- (1) This function is used to send the following data groups, beginning with the Header and ending with NUL.

Send Data	Hexadecimal	Decimal	Data length
Header	37H	55	1 byte
Identifier	30H	48	1 byte
Entire Capacity (*1)	30H to 39H	48 to 57	1 to 8 bytes
NULL	00H	0	1 byte

(*1) The entire capacity is the total byte count for that domain.
 The decimal value for the entire capacity is converted to text data and sent starting from the high order end.

Example:

If the entire capacity is 1200 bytes, the "1200" (expressed hexadecimal as 31H, 32H, 30H, and 30H, decimally as 49, 50, 48, and 48) is converted to 4-byte data

- (2) When this function is used to send the entire capacity, the entire byte capacity of the domain is sent, regardless of definitions currently entered for NV Graphics data.
 (3) NV graphics function can't be used when the value for the entire capacity is '0'

6.74 Print the graphics data in the print buffer (Function 50)

Function	Prints the buffered graphics data stored in print buffer.
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Format	[GS] (L pL pH m fn
--------	---------------------

Term	ASCII: GS (L Hex: 1Dh 28h 4Ch
------	-----------------------------------

[Parameter]

pL pH : number of bytes after pH. $0 \leq (pL + pH * 256) \leq 2$
(00h $\leq$ (pL + pH * 256) $\leq$ 02h)

Default value	pL = 2 pH = 0 m = 48 fn = 2, 50
---------------	--

Explanation

- (1) Prints the buffered graphics data stored by the process of GS (L [function 112] Which used to store in print buffer. The printer cannot print when there is no graphics data stored in the buffer.
- (2) his function cannot be used in page mode is enabled.
- (3) NV Feeds the paper the required distance when printing graphics data, regardless of line feed pitch setting entered using Set line feed command.
- (4) Moves print position to the left side of the print area after printing of graphics data is completed
- (5) The printer then enters the "beginning of the line" or "no data in print buffer" state.

6.75 Transmits the remaining capacity of the NV graphics memory (Function 51)

Function	Transmits the number of the bytes of remaining memory (unused area) in the NV graphics area		
Format	[GS] (L pL pH m fn		
Term	ASCII: GS (L Hex: 1Dh 28h 4Ch		
	[Parameter]		
	pL pH : number of bytes after pH.	$0 \leq (pL + pH * 256) \leq 2$ (00h $\leq (pL + pH * 256) \leq 02h$)	
Default value	pL = 2 pH = 0 m = 48 fn = 3, 51		
Explanation			

- (1) This function is used to send the following data groups, beginning with the Header and ending with NULL.

Send Data	Hexadecimal	Decimal	Data length
Header	37H	55	1 byte
Identifier	31H	49	1 byte
Entire Capacity (*1)	30H to 39H	48 to 57	1 to 8 bytes
NULL	00H	0	1 byte

(*1) The unused capacity is the total byte count for the unused area.
The decimal value for the unused capacity is converted to text data and sent starting from the high order end.

Example:

If the entire capacity is 1200 bytes, the "1200" (expressed hexadecimal as 31H, 32H, 30H, and 30H, decimally as 49, 50, 48, and 48) is converted to 4-byte data

6.76 Transmits the key code list for defined NV graphics (Function 51)

Function	Transmits the defined NV graphics key code list.
Format	[GS] (L pL pH m fn d1 d2
Term	ASCII: GS (L Hex: 1Dh 28h 4Ch
[Parameter]	
pL pH : number of bytes after pH.	$0 \leq (pL + pH * 256) \leq 4$ $(00h \leq (pL + pH * 256) \leq 04h)$

Default value	pL = 4 pH = 0 m = 48 fn = 64 d1 = 75 d2 = 67
---------------	---

Explanation

- (1) This function is used to send the following data groups, beginning with the Header and ending with NUL.

Send Data	Hexadecimal	Decimal	Data length
Header	37H	55	1 byte
Identifier	72H	114	1 byte
Identification status	40H	64	1 byte
Data (*1)	30H to 39H	48 to 57	2 to 80 bytes
NUL	00H	0	1 byte

(*1) The data groups are arranged according to the key codes.

When no key codes are present, the data shown below (beginning with Header and ending with NUL) is sent.

Send Data	Hexadecimal	Decimal	Data length
Header	37H	55	1 byte
Identifier	72H	114	1 byte
Identification status	40H	64	1 byte
NUL	00H	0	1 byte

6.77 Delete all NV graphics data (Function 65)

Function	Deletes all the NV graphics data.		
Format	[GS] (L pL pH m fn d1 d2 d3		
Term	ASCII: GS (L Hex: 1Dh 28h 4Ch		
	[Parameter]		
	pL pH : number of bytes after pH.	$0 \leq (pL + pH * 256) \leq 5$	$(00h \leq (pL + pH * 256) \leq 05h)$
Default value	pL = 5 pH = 0 m = 48 fn = 65 d1 = 67 d2 = 76 d3 = 82		
Explanation	(1) Ignores this command when defining macro		

6.78 Delete the specified NV graphics data (Function 66)

Function	Deletes the specified NV graphics data defined by the key codes(kc1 and kc2).		
Format	[GS] (L pL pH m fn kc1 kc2		
Term	ASCII: GS (L Hex: 1Dh 28h 4Ch		
	[Parameter]		
	pL pH : number of bytes after pH.	$0 \leq (pL + pH * 256) \leq 4$ (00h $\leq$ (pL + pH * 256) $\leq$ 04h)	
	Kc1	$32 \leq kc1 \leq 126$ (20h $\leq$ kc1 $\leq$ 7Eh)	
	Kc2	$32 \leq kc2 \leq 126$ (20h $\leq$ kc2 $\leq$ 7Eh)	
Default value	pL = 4 pH = 0 m = 48 fn = 66		
Explanation			

6.79 Define NV graphics data (raster format) (Function 67)

Function Defines the NV graphics data as a record specified by the key codes(kc1 and kc2) in the NV graphic area.

Format ① [GS] (L pL pH m fn a kc1 kc2 b xL xH yL yH [c d1...dk]1... [c d1...dk]b
② [GS] 8 L p1 p2 p3 p4 m fn a kc1 kc2 b xL xH yL yH [c d1...dk]1... [c d1...dk]b

Term ASCII: GS (L
Hex: 1Dh 28h 4Ch

[Parameter]

① pL pH : number of bytes after pH. $12 \leq (pL + pH * 256) \leq 65535$
(0Ch $\leq (pL + pH * 256) \leq$ FFFFh)

② p1 p2 p3 p4 : number of bytes after p4
 $12 \leq (p1 + p2 * 256 + p3 * 65536 + p4 * 16777216) \leq 4294967295$
0Ch $\leq (p1 + p2 * 256 + p3 * 65536 + p4 * 16777216) \leq$ FFFFFFFFh

m = 48

fn = 67

a = 48

Kc1 $32 \leq kc1 \leq 126$ (20h $\leq kc1 \leq$ 7Eh)

Kc2 $32 \leq kc2 \leq 126$ (20h $\leq kc2 \leq$ 7Eh)

b = 1 (Number of colors for the defined data)

xL, xH : number of dots in horizontal direction as (xL + xH * 256)

$1 \leq (xL + xH * 256) \leq 576$ (0 $\leq xL \leq 255$, 0 $\leq xH \leq 2$)

yL, yH : number of dots in vertical direction as (yL + yH * 256)

$1 \leq (yL + yH * 256) \leq 512$ (0 $\leq yL \leq 255$, 0 $\leq yH \leq 2$)

c = 49 (Specifies the color of the defined data)

d specifies the defined data (raster format)

$k = (\text{int} ((xL + xH * 256) + 7) / 8) * (yL + yH * 256)$

k indicates the number of the definition data.

Explanation

- (1) If the specified key code already exists in memory, it will be necessary to overwrite the data.
- (2) Ignores If there is insufficient capacity available for storing NV graphics data, Function 67 cannot be used. Use Function 51 to confirm the available capacity in the NV graphics data area
- (3) The data for the byte k of d1...dk is processed as a single item of defined NV graphics data. The defined data (d) specifies "1" for bits corresponding to dots that will be printed and "0" for bits corresponding to dots that will not be printed.
- (4) NV graphics data is printed using Function 69.
- (5) The relationship between NV graphics data (raster format) and print results is shown in the table below.

d1	d2	...	dx
dx+1	dx+2	...	dx+2
:	:	...	:
...	dk-2	dk-1	dk

MSB LSB MSB LSB MSB LSB MSB LSB
X = (xL + xH * 256)

- (6) Ignores this command when defining macro.

6.80 Print the specified NV graphics data (raster format) (Function 69)

Function Prints the NV graphic data defined by the key codes (kc1 and kc2).

Format [GS] (L pL pH m fn kc1 kc2 x y

Term ASCII: GS (L
Hex: 1Dh 28h 4Ch

[Parameter]

pL pH : number of bytes after pH. $0 \leq (pL + pH * 256) \leq 6$
(00h $\leq (pL + pH * 256) \leq 06h$)

pL = 6
pH = 0
m = 48
fn = 69
kc1 : $32 \leq kc1 \leq 126$
kc2 : $32 \leq kc2 \leq 126$
x = 1, 2
y = 1, 2

Explanation

- (1) This function is used to print NV graphics data defined using Function 7 of this command.
- (2) The printer only prints when NV graphics data corresponding to the specified key code is present.
- (3) Users have the option of specifying horizontal (times x) x vertical (times y) Size settings for the selected data.
- (4) his function is valid in standard mode “at the beginning of the line” or “no data in print buffer”.
- (5) This function is valid.
- (6) Graphics print will work only in rotate 0. Other than rotate 0, unexpected results will occur while printing.

Scaling	Vertical		Horizontal	
(x, y)	Dot Density	Max. Print area (dot)	Dot Density	Max. Print area (dot)
		Paper width 80mm		Paper width 80mm
1	203 dpi	512	203 dpi	576
2	101 dpi	256	101 dpi	288

- (7) In page mode, printing operations will not be performed simply because image data has been stored to the printer buffer.

6.81 Store the graphics data in the print buffer (raster format) (Function 112)

Function Stores the graphics data (raster format) in the print buffer.

Format ① [GS] (L pL pH m fn a bx by c xL xH yL yH d1...dk
② [GS] 8 L p1 p2 p3 p4 m fn a bx by c xL xH yL yH d1...dk

Term ASCII: GS (L
Hex: 1Dh 28h 4Ch

[Parameter]

① pL pH : number of bytes after pH. $11 \leq (pL + pH * 256) \leq 65535$
(0Bh $\leq (pL + pH * 256) \leq$ FFFFh)

② p1 p2 p3 p4 : number of bytes after p4

$11 \leq (p1 + p2 * 256 + p3 * 65536 + p4 * 16777216) \leq 4294967295$

0Bh $\leq (p1 + p2 * 256 + p3 * 65536 + p4 * 16777216) \leq$ FFFFFFFFh

m = 48

fn = 112

a = 48

bx = 1, 2

by = 1, 2

c = 49; When single –color printing control is selected.

xL, xH : xL and xH specify the number of dots in the horizontal direction as

$(xL + xH * 256). 1 \leq (xL + xH * 256) \leq 576 ; (0 \leq xL \leq$

255, $0 \leq xH \leq 2)$

yL, yH : yL and yH specify the number of dots in the horizontal direction as

$(yL + yH * 256). 1 \leq (yL + yH * 256) \leq 512 ; (0 \leq yL \leq$

255, $0 \leq yH \leq 2)$

d specifies the defined data (raster format)

$k = (\text{int} ((xL + xH * 256) + 7) / 8) * (yL + yH * 256)$

k indicates the number of the definition data.

Explanation

(1) The graphics data is enlarges by bx and by in the horizontal and

Vertical directions.

bx, by	Vertical direction	Horizontal direction
1	203dpi	203dpi
2	101dpi	101dpi

(2)This function is valid in standard mode “at the beginning of the line” or “no data in print buffer”.

(3)The data for the byte k of d1...dk is processed as a single item of defined NV graphics data. The defined data (d) specifies “1” for bits corresponding to dots that will be printed and “0” for bits corresponding to dots that will not be printed.

(4)NV graphics data that exceeds the print area for one line will not be printed.

(5)Graphics print will work only in rotate 0. Other than rotate 0, unexpected results will occur while printing.

(6)The relationship data between NV graphics data (raster format) and print results is shown in the table below.

d1	d2	...	dx
dx+1	dx+2	...	dx+2
:	:	...	:
...	dk-2	dk-1	dk

MSB LSB MSB LSB MSB LSB MSB LSB
X = $(xL + xH * 256)$

(7)In page mode, printing operations will not be performed simply because image data has been stored to the printer buffer.

8.10.71 TRANSMIT PRINTER ID, REMOTE DIAGNOSTICS EXTENSION

[GS]I@

Function	Performs the remote diagnostic function specified by <i>n</i>
----------	---

Format	[GS] I @ <i>n</i>
--------	-------------------

Term	ASCII: GS I @ Hex: 1Dh 49h 40h
------	-----------------------------------

[Parameter]

n : 32 – 255 (not all defined but reserved)

Explanation

- (1) Most of the diagnostic items are maintained in non-volatile memory (NVRAM), but some are maintained in read-only memory (ROM).
- (2) The printer returns all data in ASCII format. It is preceded by the parameter *n* to identify the diagnostic item and is followed by a Carriage Return (0D) to signify the end of the data.
- (3) The command performs the remote diagnostic function specified by *n* as described in the following table

n (hex)	Remote diagnostic item	Function
23	Serial Number	Return serial number, a total of 13 bytes
27	Model Number	Return Model number, a total of 19 bytes
2F	Boot firmware CRC 4 digit ASCII	Return boot firmware CRC, a total of 6 bytes
37	Flash firmware CRC 4 digit ASCII	Return flash firmware CRC, a total of 6 bytes.
97	Boot firmware version	Return boot firmware version, returns 6 bytes
A3	Flash firmware version	Return flash firmware version, returns 6 bytes

6.82 Transmit status [GS] r

Function Transmits the printer status specified by n

Format [GS] r n

Term ASCII: GS r n
Hex: 1Dh 72h 40h

[Parameter]

n : 1, 49 – Printer Status

2, 50 – Cash drawer status

Explanation

- (1) This is a batch mode command which transmits the response after all prior data in the receive buffer has been processed. There may be a time lag between the printer receiving this command and transmitting the response, depending on the receive buffer status
- (2) The status bytes to be transmitted are described in the following tables.

Printer status (n=1, 49)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Paper present
	On	01	1	Paper exhausted
1	Off	00	0	Cover closed
	On	02	2	Cover open
2	Off	00	0	Paper present
	On	04	4	Paper exhausted
3	-	-	-	Undefined
4	Off	00	0	Not used. Fixed to off.
5	-	-	-	Undefined
6	-	-	-	Undefined
7	Off	00	0	Not used. Fixed to off.

Cash drawer status (n=2, 50)

No cash drawer support in B-FP3D –GH/GS printer. So Always returns 0x03

- (3) When n is out of the specified range, the command is ignored.

6.83 Real time status transmission

[ESC] N

Function	Transmits the selected one byte printer status specified by n in real time
Format	[ESC] n n
Term	ASCII: ESC n Hex: 1Bh 6Eh

[Parameter]

n : 1 = ALL status (4bytes status)

Explanation	(1) The command is ignored if n is out of range. (2) The status bytes to be transmitted are described in the following four tables.
-------------	--

Transmit printer status (1'st byte of 4bytes)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
2	On	04	4	Undefined. Fixed to on
3	Off	00	0	On line state (Not busy)
	On	08	8	Off line state (Busy)
4	On	10	16	Not used. Fixed to on.
5	-	-	-	Undefined
6	-	-	-	Undefined
7	Off	00	0	Not used. Fixed to off.

Transmit Offline status (2'nd byte of 4bytes)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
2	Off	00	0	Cover closed
	On	04	4	Cover open
3	Off	00	0	Paper Feed Button is not pressed
	On	08	8	Paper Feed Button is pressed
4	On	10	16	Not used. Fixed to on.
5	Off	00	0	Printing not stopped due to paper condition
	On	20	32	Printing stopped due to paper condition
6	Off	00	0	No error condition
	On	40	64	Error condition exists in the printer
7	Off	00	0	Not used. Fixed to off.

Transmit error status (3'rd byte of 4bytes)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
2	Off	00	0	Undefined. Fixed to off
3	Off	00	0	Undefined. Fixed to off
4	On	10	16	Not used. Fixed to on.
5	Off	00	0	No unrecoverable error
	On	20	32	Unrecoverable error occurred
6	Off	00	0	Thermal print-head temp. and power supply voltage are in range

	On	40	64	Thermal print-head temp. and power supply voltage are out of range
7	Off	00	0	Not used. Fixed to off.

Transmit paper roll sensor status (4'th byte of 4bytes)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
2	Off	00	0	Undefined. Fixed to off.
3	Off	00	0	Undefined. Fixed to off.
4	On	10	16	Not used. Fixed to on.
5	Off	00	0	Paper present
	On	20	32	Paper exhausted
6	Off	00	0	Paper present
	On	40	64	Paper exhausted
7	Off	00	0	Not used. Fixed to off.

6.84 real time printer status transmission

[ESC] H

Function	Transmits the one byte status of the printer in real time.		
Format	[ESC] h		
Term	ASCII: ESC h Hex: 1Bh 68h		
Explanation	(1) Transmits one byte status of the printer in real time. (2) The status bytes to be transmitted are described in the following table		

Value of Byte:

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	Off	00	0	Undefined. Fixed to off.
2	Off	00	0	Cover closed
	On	04	4	Cover open
3	Off	00	0	Undefined. Fixed to off.
4	Off	00	0	One or both cash drawers open
	On	10	16	Both cash drawers closed
5	Off	00	0	Undefined. Fixed to off.
6	Off	00	0	No error condition
	On	40	64	Error condition exists in the printer
7	On	00	1	Not used. Fixed to on.

8.10.75 REAL TIME STATUS TRANSMISSION

[DLE] EOT

Function Transmits the selected one byte printer status specified by *n* in Real Time.

Format [DLE] EOT *n*

Term ASCII: DLE EOT
Hex: 10h 04h

[Parameter]

- n* : 1 = Transmit printer status
2 = Transmit offline status
3 = Transmit error status
4 = Transmit paper sensor status

Explanation (1) The command is ignored if *n* is out of range
(2) The status bytes to be transmitted are described in

the following tables.

Transmit printer status (n = 1)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
	On	04	4	Undefined. Fixed to on
3	Off	00	0	On line state (Not busy)
	On	08	8	Off line state (Busy)
4	On	10	16	Not used. Fixed to on.
5	-	-	-	Undefined
6	-	-	-	Undefined
7	Off	00	0	Not used. Fixed to off.

Transmit Offline status (n = 2)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
2	Off	00	0	Cover closed
	On	04	4	Cover open
3	Off	00	0	Paper Feed Button is not pressed
	On	08	8	Paper Feed Button is pressed
4	On	10	16	Not used. Fixed to on.
5	Off	00	0	Printing not stopped due to paper condition
	On	20	32	Printing stopped due to paper condition
6	Off	00	0	No error condition
	On	40	64	Error condition exists in the printer
7	Off	00	0	Not used. Fixed to off.

Transmit error status (n = 3)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
2	Off	00	0	Undefined. Fixed to off
3	Off	00	0	Undefined. Fixed to off.
4	On	10	16	Not used. Fixed to on.
5	Off	00	0	No unrecoverable error
	On	20	32	Unrecoverable error occurred
6	Off	00	0	Thermal print-head temp. and power supply voltage are in range

	On	40	64	Thermal print-head temp. and power supply voltage are out of range
7	Off	00	0	Not used. Fixed to off.

Transmit paper roll sensor status (n = 4)

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Undefined. Fixed to off.
1	On	02	2	Undefined. Fixed to on
2	Off	00	0	Undefined. Fixed to off.
3	Off	00	0	Undefined. Fixed to off.
4	On	10	16	Not used. Fixed to on.
5	Off	00	0	Paper present
	On	20	32	Paper exhausted
6	Off	00	0	Paper present
	On	40	64	Paper exhausted
7	Off	00	0	Not used. Fixed to off.

6.85 select pitch for hri characters [GS] f(Not supported)

Function	Selects standard or compressed font for printing Barcode characters		
Format	[GS] f n		
Term	ASCII: GS f Hex: 1Dh 66h n [Parameter] n : Pitch 0, 48 = Standard Pitch at 16.9 CPI 1, 49 = Compressed Pitch at 22.6 CPI 0, 48 = Standard Pitch at 15.6 CPI 1, 49 = Compressed Pitch at 20.3 CPI		
Default value	n : 0		
Explanation	(1)	Selects standard or compressed font for printing Bar Code characters.	
	(2)	The command is ignored if n is out of range.	

6.86 SELECT MEMORY TYPE WHERE TO SAVE LOGOS OR USER DEFINED FONTS OR MACROS [US] 3

Function	Specifies whether to load the logos or user-defined characters or macro to Flash Memory or to RAM
Format	[US] 3 n
Term	ASCII: US 3 Hex: 1Fh 33h [Parameter] n : 48 = Loads active logo to RAM only n : 49 = Loads active logo to Flash memory n : 50 = Loads user-defined characters to RAM only n : 51 = Loads user-defined characters to Flash memory n : 52 = Loads macro to RAM only n : 53 = Loads macro to Flash memory
Reference	Define Download bit image (1Dh 2Ah) Start or End Macro definition (1Dh 3Ah) Execute Macro (1Dh 5Eh) Download BMP logo 1Fh (+*.BMP)

6.87 FLASH ALLOCATION [US] 4

Function	This command sets the allocation of Flash sectors between user data storage and logos / user-defined characters
----------	---

Format	[US] 34 n1 n2
--------	---------------

Term	ASCII: US 4 Hex: 1Fh 34h
------	-----------------------------

[Parameter]

n1 : Number of 64k sectors used for logos and user-defined characters

n2 : Number of 64k sectors used for user data storage

$n1 + n2 \leq 5$

Default value	n1 : 1 n2 : 1
---------------	---------------

Explanation	(1) This command sets the allocation of Flash sectors between user data storage and logos / user defined characters. This allocation is saved in the FLASH of the printer and is therefore saved across power cycles. Printer always keeps 64 Kbytes for User-defined characters.
-------------	---

n1	n2	User Defined Character	Download Logo	User Storage Data
0	5	64 Kbytes	0 Kbytes	1 Kbytes
1	4	64 Kbytes	64 Kbytes	1 Kbytes
2	3	64 Kbytes	128 Kbytes	1 Kbytes
3	2	64 Kbytes	192 Kbytes	1 Kbytes
4	1	64 Kbytes	256 Kbytes	1 Kbytes
5	0	64 Kbytes	320 Kbytes	0 Kbytes

(2) If $n1 + n2$ is greater than the maximum number of sectors available, the command is ignored.

(3) Reissuing this command with different parameters will erase all sectors.

6.88 Erase user FLASH sector [US] 5

Function	Erases a page of Flash Memory and sends a carriage return when the operation is complete
Format	[US] 35 n
Term	ASCII: US 5 Hex: 1Fh 35h [Parameter] n : 49

Explanation

- (1) This command erases all sectors available for user-defined characters and multiple Logos.
- (2) When the logo definition area is full and an application is attempting to define new Logos, and when an application wants to replace one user-define character set with Another. In both cases, all logos and character set definitions are erased and must be redefined
- (3) After sending the Erase user flash sector (1F 35 n) command, an application should Wait for response from printer (0x0D) before sending the data. Otherwise data will be lost.

6.89 Write to user data storage [FS] g

Function	Stores the data to flash memory
Format	[FS] g 1 m a1 a2 a3 a4 nL nH d1...dk
Term	ASCII: FS g 1 Hex: 1Ch 67h 31h [Parameter] m : 0 a : $0 \leq (a1 + a2 * 256 + a3 * 65536 + a4 * 16777216) \leq 1023$ nL nH : $1 \leq (nL + nH * 256) \leq 1024$; $(0 \leq (nL \leq 255) (0 \leq (nH \leq 04)$ d : $32 \leq d \leq 255$ k : $(nL + nH * 256)$

Explanation

- (1) Stores the data to Flash memory.
- (2) a1, a2, a3 and a4 specifies the number of stored data bytes $(nL + nH * 256)$
- (3) d specifies the stored data.
- (4) k indicates the number of stored data. K is an explanation parameter. Therefore, it does not need to be transmitted.
- (5) This command is effective only when processing at the head of the line.
- (6) This command is invalid in page mode.
- (7) In the following cases, This command becomes invalid and usually processes since the next data as data

Note

When parameter (m) is a definition beyond the border.

When the storage beginning address (a1, a2, a3 and a4) is definition beyond the border.

When number (nL, nH) of storage data is a definition beyond the border.

When the storage beginning address (a1, a2, a3 & a4) + number (nL, nH) of storage data ≥ 1024 .

6.90 Read FROM User Data Storage [FS] g

Function	Transmits the data in flash memory from specified address.
----------	--

Format	[FS] g 2 m a1 a2 a3 a4 nL nH
--------	------------------------------

Term	ASCII: FS g Hex: 1Ch 67h 32h
------	---------------------------------

[Parameter]

m : 0

a : $0 \leq (a1 + a2 * 256 + a3 * 65536 + a4 * 16777216) \leq 1023$
 $(0 \leq a1 \leq 255, 0 \leq a2 \leq 3, a3=0, a4 = 0)$

nL nH : $1 \leq (nL + nH * 256) \leq 1024; (0 \leq (nL \leq 255) (0 \leq (nH \leq 04))$

Explanation

(1) Transmits the data in Flash memory to host.

(2) a1, a2, a3 and a4 specifies the starting address of transmission data as
 $(a1 + a2 * 256 + a3 * 65536 + a4 * 16777216)$

(3) nL and nH specifies the amount of transmission data bytes as $(nL + nH * 256)$

bytes.

(4) Transmission data [Header + data + NULL] as follows is transmitted:

<u>Transmitted data</u>	<u>Hex</u>	<u>Decimal</u>
<u>Amount of data</u>		
5FH	95	Header 1 byte
32~255	(nL + nH x 256) bytes	Flash memory data 20H~FFH
00H 1 byte	0	NULL

6.91 PRINTER SETTING CHANGE [US] 6

Function Set the printer configuration specified by m, n

Format [US] 6 [m n], [m n], ..., [m n] 0FFH

Term ASCII: US 6
Hex: 1Fh 36h

[Parameter]

m : setting parameter type

n : parameter value

Explanation (1) Printer setting parameter and parameter value as per below table

M (Hexadecimal)	Function	n (Hexadecimal)	Function
21	Default lines per inch (Only Font Size 1 is effective)	00 01 02	8.13 lines per inch 7.52 lines per inch * 6 lines per inch
22	Carriage return usage	00 01	Ignore CR * Use CR as Print cmd. *
25	Receipt synchronization	00 01	Synchronization enabled Synchronization disabled *
27	PDF417 Print Column	00 01	9 Columns * 14 Columns
40	Default code page	00 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18	437 * 850 852 858 860 Not Defined 863 Not Defined 865 866 1252 Not Defined Not Defined Not Defined Not Defined 857 1253 1254 1256
70	DBCS/FontSize	00 01 02 03	Invalid/Size1* Invalid/Size2 Valid/Size1 Valid/Size2
71	DBCS Print Mode	00 01	Valid* Invalid
72	Additional Feed	00 01 02 03 04 05 06 07	Disable* 3mm 6mm 9mm 12mm 15mm 18mm 21mm

73	DBCS Select CG	00	Japanese
		01	Chinese
		02	Korean

Set the printer configuration specified by m and n. If m or n is out of range, this command is ignored.
But the printer waits the data until terminator code "0FFH".

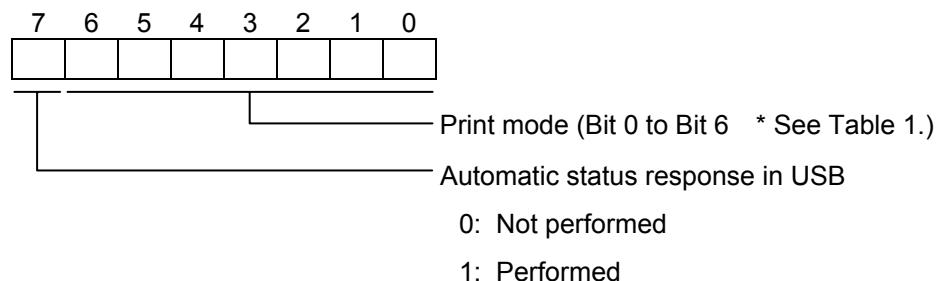
7. TEC ORIGINAL COMMAND

7.1 Mode select command [ESC] M

Function Changes the print mode.

Format [ESC] M; a (,b) [LF] [NUL]

Term a: Print mode designation



* Table 1 Print mode

HEX	Mode	How to deal with the received data after an error is cleared
30H	LABEL	Discards
31H	ESC/POS	Continues printing
32H	ESC/POS	Continues printing
34H	ESC/POS	Continues printing
41H	TPCL	Continues printing
42H	TPCL1	Continues printing
43H	C Mode	Continues printing
44H	Z Mode	Continues printing

b: Print position detection feed (Omissible. If omitted, the print position detection feed is not performed.)

0: When the mode change from ESC/POS to other Mode is requested, a print position detection feed is not performed after the mode is changed.

1: When the mode change from ESC/POS to other Mode is requested, a print position detection feed is performed after the mode is changed.

7.2 Reset command [ESC] WR

Function	Returns the printer to its initial state.
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Format	[ESC] WR [LF] [NUL]
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Explanation

- (1) The printer is returned to the same state as when the power is turned on.
- (2) When the printer receives this command during printing, it is initialized after issuing the label which is being printed.
- (3) After the Initialize Command is sent (or after printing is completed, if printing is performed), the next command must not be sent within approximately 30 seconds on the wireless LAN model or within approximately 5 seconds on other models, because the printer is initialized.
- (4) When receiving this command during data transmission, the printer is initialized after completing the transmission.

Notes

- (1) If a command error or communication error occurs when receiving the Reset Command, an error message is displayed in the online mode. However, it is not displayed in the SYSTEM mode.

Example	[ESC] WR [LF] [NUL]
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7.3 Status request command [ESC] v, [ESC] FM, [ESC] WS

Function Requests that the printer sends back the printer status and the battery status to the host.

Format [ESC] v
[ESC] FM [LF] [NUL]
[ESC] WS [LF] [NUL]

Explanation (1) When this command is received, the printer sends the printer status and battery status to the host.

STX	Printer ID		Printer status	Battery status
02H	xxH	xxH	xxH	xxH

Printer ID 2-byte hex data (in order from High to Low)

Printer status Printer status is indicated in 1-byte data.

- "00": Normal state
- "01": Cover open state
- "02": Operating
- "03": Possession by other terminal
- "04": Pause state
- "05": Wait for strip
- "06": Command syntax error
- "09": Label end
- "11": Paper jam
- "13": Label end
- "15": Cover open error
- "17": Broken head dots error
- "18": Thermal head excessive temperature
- "19": Ambient temperature error
- "32": Abnormal battery temperature
- "33": Battery excessive temperature
- "36": Low battery
- "37": Charging error
- "39": Bluetooth setup error (including initialization error)
- ("40": Normal issue end) Response status for automatic status transmission
- ("41": Normal feed end) Response status for automatic status transmission
- "45": Wait for battery recovery
- "46": Wait for head temperature reduction
- "47": Wait for motor temperature reduction
- "50": Flash ROM write error
- "51": Flash ROM erase error
- "54": Flash ROM storage area full state
- "55": Writable character/PC command save mode

NOTE: 0DH (Normal issue end + Label end) is a state when the printer runs out of labels, after a label is issued.

Battery status

The battery charge status is indicated in 5 levels.

- 01H: 7.2V or less
- 02H: 7.3V to 7.4V
- 03H: 7.5V to 7.7V
- 04H: 7.8V to 7.9V
- 05H: 8.0V to 8.4V

7.4 Mode Information Acquire Command [ESC] WX

Function Sends the printer mode information to the host.

Format [ESC] WX [LF] [NUL]

Explanation

(1) The mode information format to be sent to the host, is as follows:

STX	Mode information (16 bytes)															
	E	S	C	/	P	O	S	SP	SP	SP	SP	SP	SP	SP	SP	SP
02H	45H	53H	43H	2FH	50H	4FH	53H	20H	20H	20H	20H	20H	20H	20H	20H	20H

Example [ESC] WX [LF] [NUL]